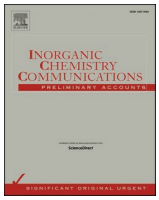
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Electrochemical sensing of pharmaceutical pollutants using modified glassy carbon electrodes with nanostructures: A review

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ABSTRACT

Pharmaceutical pollutants pose a significant threa[t] [to] environmental a[nd] [public] health due to their persistence and bioaccumulation. Electrochemical sensing [has] e[mer]ged as a power[ful] technique for detecting these contaminants with high sensitivity, selectivity, and rapid [re]sponse times. Among various electrode materials, modified glassy carbon electrodes (GCEs) [in]corporating nan[ostruc]tures have gained considerable attention for their enhanced electrochemical perf[ormance]. This review pr[ovide]s a comprehensive overview of recent advancements in the electrochemical [sens]ing of pharmaceutical pollutants using nanostructure-modified GCEs. We discuss different types of nanomat[erial]s, including [metal] nanoparticles, carbon-based nanostructures, and metal–organic frameworks, highlightin[g] [the]ir roles in i[mpro]ving electron transfer, surface area, and analyte interaction. Furthermore, we explore th[e] [ana]lytical p[erfo]rmance of these sensors, challenges in their practical application, and futur[e] [per]spectives [for] [the] [develo]pment of highly efficient electrochemical platforms for pharmaceutical pollu[tant] [detect]ion.

1. Introduction

Pharmaceutical pollutants have emerged as [a] [signific]ant environmental and public health concern due to their pers[istence] [in] wat[er] and food sources [1,2]. These pollutants [ori]ginate [from] both human excretion and improper disposal of unu[sed] drugs, leading [to] unintended exposure through contaminated dri[nking] [wa]ter and food [3]. While the concentrations in tap water are often extre[mely] low, their long-term effects remain uncertain, esp[ecial]ly given the c[halle]nges in detection and risk assessment [4]. W[astew]ater treatment p[lant]s are often ineffective in entirely rem[ov]ing the[se] compounds, contributing to the spread of antibiotic-resis[tant] b[acte]ria and [gene]s, which pose a severe threat to both hum[an] healt[h] and ecosy[stem]s [5,6]. Additionally, the increasing glo[bal] consumptio[n] [of] [pharm]aceuticals, driven by aging populations [and] risi[ng] healthca[re] [acc]ess, is expected to exacerbate environm[ental] cont[amin]ation [7]. As research shifts towards risk assessment [and] mitig[ation], there is a growing urgency to develop greener produ[ction] processes, enhance drug disposal methods, and implement advanc[ed] water treatment technologies to curb the spread of pharmaceutical resid[ues] [8].

Detecting pharmaceutical pollutants in water is crucial due to their diverse nature and biological effects, even at low concentrations [9,10]. Unlike traditional contaminants, pharmaceuticals require highly sensitive detection methods to ensure accurate monitoring [11]. The absence of comprehensive regulatory frameworks further underscores the need for robust detection tools. Advanced sensor technologies are transforming water quality monitoring by enabling real-time detection, facilitating the prompt identification of contamination sources, and allowing continuous water quality assessment. This proactive approach supports adaptive water management strategies, ensuring safer water resources and mitigating potential environmental and public health risks [12].

Electrochemical sensing techniques have gained significant attention for detecting pharmaceutical pollutants due to their high sensitivity, reliability, and practical application [13–15]. Unlike traditional methods, electrochemical sensors simplify electrode modification challenges, allowing efficient recognition and amplification reactions in homogeneous solutions [16,17]. This enhances their ability to detect various analytes with maximum efficiency [18,19]. Additionally, their cost-effectiveness and straightforward operation make them a widely accepted alternative for water quality monitoring [20]. As research continues, the development of advanced electrochemical platforms is expanding their applications, offering a powerful and promising approach for real-time environmental analysis [21,22].

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GCEs play a crucial role in electrochemical sensing due to their excellent mechanical, electrical, and chemical properties. Produced through the carbonization of phenol–formaldehyde resin, GCEs possess a dense, non-graphitic structure that ensures high stability and durability. Their low electrical resistivity and compatibility with miniaturization make them ideal for various applications, including biosensing, cell detection, and cancer biomarker analysis [23–27]. Nanotechnology involves the manipulation of materials at the nanoscale, typically between 1 and 100 nm [28–32]. In recent years, nanomaterials have found widespread applications in the fields of chemistry, physics, mathematics, physics, and basic sciences in general [33–37]. At this scale, materials often exhibit unique physical and chemical properties not seen in their bulk counterparts [38–42]. These include increased surface area, enhanced reactivity, and improved mechanical, electrical, and thermal behavior [38,43–46]. As a result, nanotechnology offers significant advantages across many scientific and industrial domains [9,47,48]. It is widely used in drug delivery systems, corrosion protection, water purification technologies, and energy storage solutions [49–51]. In the context of electrochemical sensors, nanomaterials—such as metal nanoparticles, carbon nanostructures, and metal–organic frameworks (MOFs)—can be used to modify GCEs, greatly enhancing their sensitivity, selectivity, and reproducibility [52]. This makes nanostructure-functionalized GCEs highly effective for real-time, in situ electrochemical analysis across a range of scientific and industrial applications [53–57].

This review paper focuses on the application of electrochemical sensing techniques for the detection of pharmaceutical pollutants, with particular emphasis on GCEs. It provides a comprehensive overview of the principles of electrochemical detection, key performance parameters, and the critical role of electrode surface modification in enhancing sensor performance. The novelty of this review lies in its in-depth exploration of recent advances in GCE modifications—especially the incorporation of metal nanoparticles and carbon-based nanostructures—to improve the sensitivity, selectivity, and stability of electrochemical sensors. Unlike previous reviews that broadly examine carbon-based electrodes or general electrode modification strategies, this review focuses explicitly on GCE-specific challenges and solutions, offering a comprehensive comparison of classical and emerging modification techniques (such as soft plasma polymerization and laser ablation) tailored to GCE performance optimization. By focusing on these cutting-edge approaches, the paper offers fresh insights into the design of high-performance electrochemical sensors for environmental monitoring. Furthermore, it outlines emerging research directions and underscores the transformative potential of advanced GCE modifications in addressing current challenges in pollutant detection.

2. Pharmaceutical pollutants and the need to detect them in the environment

Pharmaceutical pollutants have become a growing environmental concern due to their continuous introduction into aquatic ecosystems and their potential for long-term ecological and human health effects [58–60]. Pharmaceutical chemicals that enter water bodies, including surface and groundwater, originate from a variety of sources (Fig. 1) [61]. The primary sources of these pollutants are anthropogenic activities, particularly human and veterinary pharmaceutical use. After consumption, a significant portion of active pharmaceutical ingredients (APIs) is excreted as metabolites or unchanged compounds in urine and feces, entering municipal wastewater treatment plants (WWTPs). However, conventional WWTPs are not explicitly designed to remove pharmaceuticals, allowing many drugs to pass through and enter natural water bodies [62]. In addition to excretion, improper disposal of unused or expired medications contributes to environmental contamination, as pharmaceuticals are often flushed down toilets or discarded in landfills, where they may leach into groundwater [63]. Furthermore, pharmaceutical manufacturing plants release effluents containing high concentrations of APIs, particularly in regions with lax regulations and inadequate treatment infrastructure [64]. Agricultural activities also contribute notably, as veterinary pharmaceuticals administered to livestock, such as antibiotics and antiparasitics, can be excreted into soil and water

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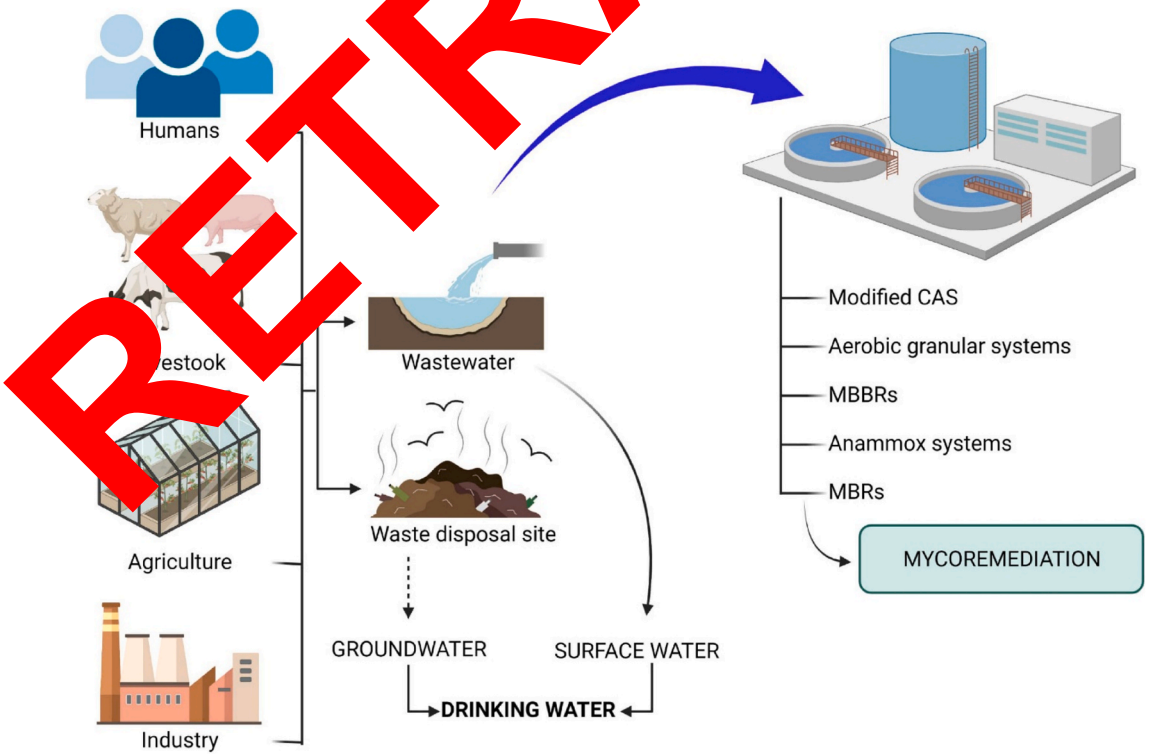


Fig. 1. Pharmaceuticals enter a body of water via bioremediation technology. (→) Direct contamination. (⇢) Contamination occurs in multiple steps. Monitoring indicates that contamination accumulates in both surface and groundwater [61].

through runoff from farms [65,66]. The increasing global consumption of pharmaceuticals, driven by population growth, aging demographics, and expanded access to healthcare, suggests that pharmaceutical pollution will continue to escalate unless effective mitigation strategies are implemented [67–69].

The detection of pharmaceutical pollutants in aquatic environments has been facilitated by advances in analytical chemistry, particularly through techniques such as high-performance liquid chromatography (HPLC), liquid chromatography-mass spectrometry (LC-MS), and electrochemical sensors. These methods enable the detection of pharmaceuticals at extremely low concentrations, typically in the nanogram per liter (ng/L) to microgram per liter (μg/L) range [70–72]. Studies have documented the presence of a wide range of pharmaceuticals in surface waters, groundwater, and even drinking water sources worldwide. Commonly detected compounds include antibiotics (e.g., sulfamethoxazole, ciprofloxacin), analgesics and anti-inflammatory drugs (e.g., ibuprofen, diclofenac), psychiatric medications (e.g., fluoxetine, diazepam), beta-blockers (e.g., propranolol), and endocrine-disrupting compounds such as synthetic estrogens (e.g., ethinylestradiol). The persistence of these pharmaceuticals in aquatic environments depends on their chemical properties, such as solubility, biodegradability, and tendency to bioaccumulate [61]. While some pharmaceuticals degrade relatively quickly under sunlight or microbial activity, others persist for extended periods, raising concerns about chronic exposure and cumulative ecological effects [73–75]. Emerging detection technologies, including biosensors, electrochemical sensors, and advanced spectroscopic techniques, offer promising tools for real-time and in situ monitoring of pharmaceutical contaminants. Such advancements are essential for understanding the spatial and temporal distribution of pharmaceuticals in the environment and informing regulatory measures [76,77].

The ecological impacts of pharmaceutical pollutants in aquat vironments are diverse and complex, affecting organisms at var trophic levels. One of the most concerning effects is the proliferation antibiotic resistance due to the continuous release of thal ant biotic concentrations into water bodies. These cond s p ote the selection and propagation of antibiotic-resistan cteria B) and antibiotic resistance genes (ARGs), which can b sfer and animal pathogens, posing significant ic h sks [78]. addition, endocrine-disrupting pharmac als, part ly synthetic estrogens from contraceptives, have b wn to induc inization of male fish, leading to reproducti bnor ties and pop ation declines in aquatic species. Such d uptions can ecosystem dynamics and reduce biodiversity [79] ychiatric drugs, ding selective serotonin reuptake inhibit SSRIs) nd benzodia epines, affect fish behavior, impairing pr r–pre eractions, foraging efficiency, and social behaviors, which ma scading fects on food web stability [80]. Nonsteroid i-infla tory d (NSAIDs), such as diclofenac, have be ike organ in fish and amphibians, with studies sho g kidn ailure an ll tissue damage at environmentally relev conc 81]. Furthermore, chronic exposure to pharmaceuti tures can sult in additive or synergistic effects, where the comb impact of multiple drugs is greater than their individual effects. Th ises concerns about long-term, low-dose exposure and the potentia or unforeseen ecological consequences [82].

Addressing pharmaceutical pollution in aquatic environments requires a multi-pronged approach involving improved wastewater treatment technologies, stricter regulatory frameworks, and enhanced public awareness. Advanced oxidation processes (AOPs), such as ozonation, photocatalysis, and membrane filtration, have shown promise in degrading pharmaceutical contaminants more effectively than conventional wastewater treatment [83]. However, the widespread implementation of these technologies faces economic and logistical challenges. Regulations governing pharmaceutical disposal need to be reinforced, with take-back programs and environmentally friendly disposal guidelines becoming more prevalent [84]. The development of green pharmaceuticals, designed to degrade rapidly in the environment while maintaining therapeutic efficacy, represents a forward-thinking solution to reduce persistent contamination. Additionally, interdisciplinary collaborations between scientists, policymakers, industry stakeholders, and the public are crucial for developing sustainable strategies to mitigate the impact of pharmaceuticals on aquatic ecosystems [85,86]. Continued research into the long-term effects of pharmaceutical pollutants, their interactions with other environmental stressors, and their potential human health risks is essential for ensuring water quality and ecosystem resilience in the face of escalating pharmaceutical consumption [87].

3. Fundamentals of electrochemical sen

Nanotechnology has revolutionize merous scie c and industrial fields such as chemistry, mathemat hysics, … enabling the manipulation of materials at th noscal 92]. A is scale, materials often exhibit unique ph al, chemical, gical properties that differ significantly fro eir bul unterpa 93–97]. One of the most prominent advantage na chnolog is the dramatic increase in surface area-to-v e rat ich enh es reactivity, sensitivity, and efficiency in ide range pli ns [98–102]. In medicine, nanoparticles g used for ta d drug delivery, imaging, and diagnostics, owing tments to be more effective while minimizing side effects [103–107]. vironmental science, nanomaterials play a cruci n pollutant de on, water purification, and remediation of taminated sites [108–111]. In electronics, nanoscale components ha led to faster aller, and more energy-efficient devices [112–114]. Fu rmore, na echnology is integral to the development of adv d senso including electrochemical sensors, where nanostructu is improve sensitivity, selectivity, and stability [115]. interdisciplinary nature fosters innovation across fields such as ology, energy storage (e.g., nanostructured electrodes for batteries d supercapacitors), and food safety (e.g., nano-packaging with antimicrobial properties) [116,117]. However, as promising as nanotechnology is, there are also challenges, including potential toxicity, environmental impact, and the need for standardization in production and regulation [118,119]. Nonetheless, ongoing research continues to unlock new possibilities, making nanotechnology a cornerstone of future scientific and technological advancement [120,121]. Electrochemical detection has emerged as a powerful analytical technique for identifying and quantifying contaminants, such as heavy metals, organic pollutants, and biomolecules, in diverse matrices including water, biological fluids, and soil (Fig. 2). The efficacy of these methods hinges on several key performance parameters: sensitivity, selectivity, limit of detection (LOD), linear range, and performance in real samples. These parameters collectively determine the reliability, accuracy, and practical applicability of electrochemical sensors, particularly in environmental monitoring, clinical diagnostics, and industrial safety [122]. This section provides a detailed examination of each parameter:

3.1. Sensitivity & selectivity

Sensitivity reflects the ability of an electrochemical method to detect minute changes in contaminant concentration, a critical attribute for trace-level analysis. It is quantitatively defined as the slope of the calibration curve, where the electrochemical signal (e.g., current in amperometric techniques or potential in potentiometric methods) is plotted against analyte concentration [123]. The formula is expressed as:

$$m = \frac{dS}{dC}$$

where m represents sensitivity, S is the measured signal, and C is the contaminant concentration. High sensitivity is essential for applications

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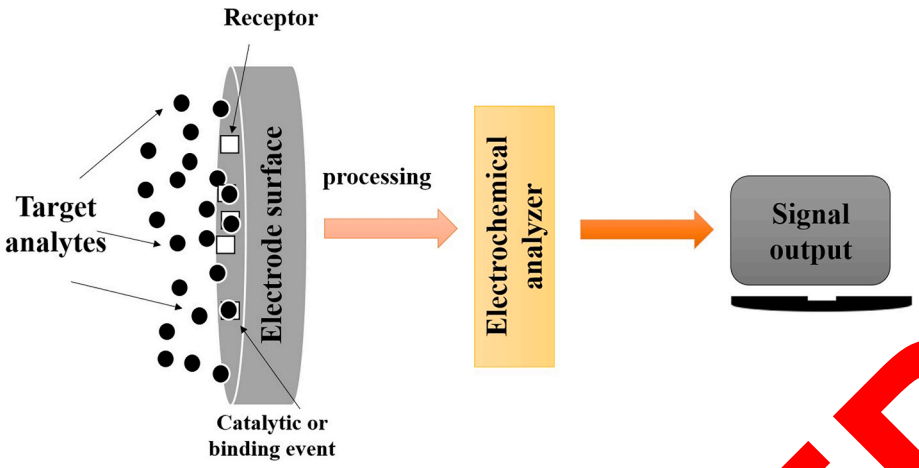


Fig. 2. The electrochemical sensor detects a target analyte through receptors on the electrode surface, triggering a ca[illegible]c or binding e[illegible]. Th[illegible]nteraction causes physicochemical changes, which are translated into electrochemical signals, varying based on the measurement t[illegible]

requiring the detection of low-level contaminants, such as heavy metals in drinking water or biomarkers in physiological fluids. Advances in nanotechnology, such as the use of nanowires or carbon nanotubes, have significantly enhanced sensitivity by improving electron transfer kinetics and increasing electrode surface area [124].

Selectivity refers to the ability of an electrochemical sensor to preferentially detect the target contaminant in the presence of potential interferents, a prerequisite for accurate analysis in complex matrices. While no universal formula exists, selectivity is often assessed by comparing the sensor's response to the target analyte against reactions to other substances, sometimes quantified as a selectivity coefficie[illegible] ion-selective electrode systems. This parameter is particularly vi[illegible] real-world samples, such as industrial effluents or blood, where mult[illegible] species coexist [125].

3.2. Limit of detection (LOD)

The limit of detection represents the l[illegible]wes[illegible] c[illegible] contaminant that can be reliably distinguis[illegible]rom b[illegible]ound noise, a cornerstone of regulatory compliance a[illegible]fety assessm[illegible] [126]. It is a crucial parameter in electrochem[illegible] s[illegible]g, influenci[illegible]e sensitivity and practical application of sensors i[illegible]ecting trace levels of substances such as heavy met[illegible] biomolecules, [illegible] pollutants. LOD is affected by factors like ele[illegible]de material, surface [illegible]odifications, electrolyte composition, a[illegible]signal [illegible]essing techniques [127]. It is calculated using the for[illegible]

$$LOD = \frac{3\sigma}{m}$$

where σ [illegible]e stan[illegible]deviation of the blank signal, and m is the sensitivity.

3.3. Linear range

The linear range defines the concentration interval over which the electrochemical signal scales linearly with contaminant concentration, facilitating straightforward quantification via a linear regression model [128]. In electrochemical detection, the linear range refers to the concentration range over which the sensor or electrode response (e.g., current, voltage, or conductance) maintains a direct proportional relationship with the analyte concentration [129]. Within this range, the calibration curve follows a linear relationship, enabling the accurate quantification of the target species. A wider linear range is desirable as it allows for the detection of both low and high analyte concentrations without requiring multiple calibration curves. However, beyond this range, the response [illegible]plate[illegible]e to ele[illegible]de saturation) or deviate (due to side reac[illegible] or mass t[illegible]ort [illegible]tations), leading to nonlinearity and red[illegible]uracy [130,[illegible]

3.4. Performance in real [illegible]les

[illegible]e ultimate test of an electrochemical sensor lies in its performance w[illegible] real sample[illegible] complex matrices like river water, blood, or industri[illegible]wastewater[illegible] here matrix effects and interferents challenge dete[illegible] accur[illegible] This parameter is evaluated through accuracy (closen[illegible]e actual value), precision (reproducibility, often [illegible]ressed as relative standard deviation, RSD), and recovery, calculated

$$Recovery = \left(\frac{Measuredconcentration}{addedconcentration}\right) \times 100\%$$

High recovery rates (e.g., 95–105 %) and low RSD values (<5%) indicate robust performance [132].

4. Types of electrodes

Electrodes play a crucial role in electrochemical analysis, with various types designed for specific applications [133]. Carbon Paste Electrodes (CPE) consist of a mixture of carbon powder and a binder, offering easy modification and high sensitivity, making them suitable for sensing applications. GCEs are composed of a complex, glass-like form of carbon with high conductivity, chemical stability, and a wide potential window, often used in voltammetry. Screen-Printed Electrodes (SPE) are disposable, cost-effective electrodes fabricated using conductive inks on plastic or ceramic substrates, widely used in portable sensing devices. Other electrodes include platinum, gold, and silver electrodes, each with unique conductivity and stability characteristics, as well as modified electrodes tailored for specific analytical applications [134].

4.1. CPE

CPEs have gained significant attention in recent years due to their unique advantages over traditional electrode materials. They offer a broader potential window, a higher signal-to-noise ratio, mechanical stability suitable for flowing systems, and resistance to passivation [135]. The latter is significant as electrode fouling is a substantial limitation in the frequent application of electroanalytical methods, especially in environmental analysis. Unlike other electrodes, CPEs are chemically inert, robust, and easily renewable [136]. They provide stable responses, low ohmic resistance, and do not require an internal

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solution, making them highly versatile for various sensing and detection applications. Additionally, CPEs are considered environmentally friendly and non-toxic, and their passivation issues can be eliminated simply by renewing their surface, which is quick and easy [137].

Since their introduction, CPEs have undergone continuous improvements. One of the most significant trends is the development of new carbon paste mixtures. Traditional graphite-based pastes are now being replaced by advanced carbon materials such as fullerenes, hollow carbon fibers, and carbon nanotubes. Simultaneously, conventional pasting liquids are being substituted with room-temperature ionic liquids (RTILs) and other newly synthesized organic compounds, improving electrode performance [136].

The fabrication of CPEs involves packing soft carbon paste into electrode bodies such as Teflon rods, glass tubes, or polyethylene syringes. These holders allow for electrical contact via a conducting wire. The electrode surface, typically between 2 mm and 10 mm in diameter, can be easily renewed by either wiping with wet filter paper or replacing the paste (Fig. 3a). This simple yet effective surface renewal ensures reproducibility comparable to more tedious polishing methods.

CPEs exhibit distinctive physico-chemical and electrochemical properties that define their behavior. They have low ohmic resistance, which makes them highly conductive, and they are inherently hydrophobic (lipophilic), affecting reaction kinetics, especially in organic redox systems. Their composite nature results in heterogeneity, and the choice of pasting liquid and carbon material can influence their stability. One limitation of CPEs is their instability in non-aqueous solutions, where some binders may dissolve over time. Additionally, aging effects are observed, primarily due to the volatility of specific binders rather than any degradation of the graphite itself.

The history of CPE development can be traced back to 1959–1963, when Adams's research group introduced them and explored their use in studying organic electrochemical reactions. By 1964–1965, modifications were introduced by incorporating organic substances into the binder, paving the way for chemically modified carbon paste electrodes (CMCPEs). A significant breakthrough occurred in 1 with the introduction of electrolytic binders, enabling studies in solid-state electrochemistry. In the 1980 s, chemically modified CPEs gained popularity, leading to significant advancements in sensor technology. This trend continued with the emergence of biologically modified CPEs in 1988, where enzymes were incorporated into the paste to create biosensors. The first glucose oxidase-based CPE was developed during this period, marking the beginning of enzyme-based electrochemical detection [138].

4.2. GCE

GCEs (Fig. 3b) have been extensively utilized in voltametric studies over the past two decades due to their unique electrochemical properties. One of their key advantages is the low residual current they exhibit over a wide potential range, particularly in aqueous media, organic solvents, and micellar solutions. This characteristic makes glassy carbon a highly effective material for indicator electrodes. However, the electrochemical performance of GCEs is susceptible to the state of the electrode surface, as the apparent rates of redox reactions at these electrodes are significantly influenced by surface activation techniques and the electrode's prior history. Consequently, much research has been devoted to understanding and refining the methods used to prepare and activate glassy carbon surfaces to ensure reproducibility and enhance electrochemical performance.

Glassy carbon was first synthesized by Yamada and Sato [139] in 1962, using phenolic resins. Its formation involves a carefully controlled heating process of phenol-formaldehyde resins in an inert atmosphere. The carbonization process begins at temperatures above 300 °C, during which oxygen and nitrogen are removed between 300 and 500 °C. This phase had to proceed gradually to allow the diffusion of gaseous by-products to the surface; otherwise, rapid heating could cause high internal pressure, leading to cracks, blisters, and distortion. As the temperature increases to the range of 500–1200 °C, hydrogen is progressively eliminated, leaving behind pure carbon. This process results in substantial volume shrinkage of about 50 %. Studies by Jenkins and Kawamura [140] using X-ray diffraction and infrared spectroscopy, researchers have revealed that glassy carbon is composed of randomly oriented aromatic ribbon-like molecules that intertwine to form a glass-like solid. Further annealing at temperatures above 1200 °C eliminates

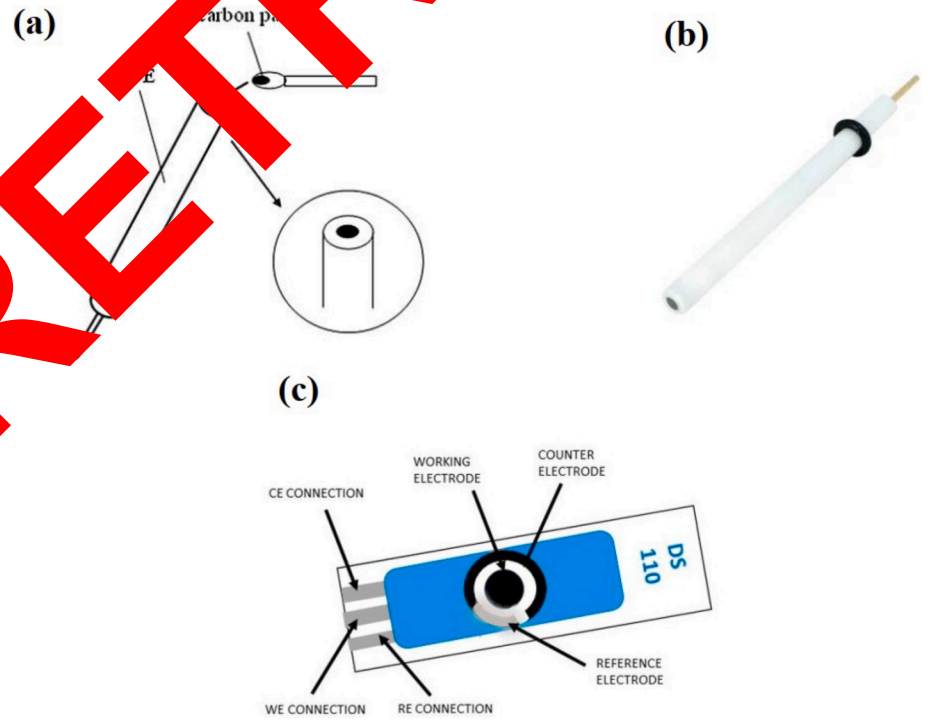


Fig. 3. (a) carbon paste, (b) glassy carbon, and (c) screen-printed electrodes.

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structural defects, producing a smooth, well-organized material with improved mechanical and electrochemical properties.

The use of glassy carbon as an electrode material in electrochemistry was first introduced by Zittel and Miller [141] in 1965. While it is widely employed for electrochemical applications, one of its primary challenges is its extreme sensitivity to surface conditions. Freshly prepared glassy carbon electrode (GCE) surfaces undergo oxidation when exposed to air and can experience irreversible changes when immersed in solutions under both open-circuit and closed-circuit conditions. The electrochemical behavior of GCEs is primarily dictated by the surface state, including the presence of functional groups, which can either facilitate or hinder electron transfer processes.

4.3. SPE

SPEs (Fig. 3c) are a key technology in the rapidly growing field of printed electronics, projected to surpass $300 billion by 2030. Their low-cost fabrication, mass production capabilities, and adaptability make them highly valuable in biosensing applications, including the detection of heavy metals (arsenite, lead, mercury), biological markers (NADH, H_2O_2, cancer biomarkers), and drug screening (codeine, diazepam). SPEs offer several advantages, such as high sensitivity, quick response, and the ability to operate at room temperature, making them ideal for in situ environmental analysis. Furthermore, their surfaces can be modified with materials like noble metals, nanomaterials, enzymes, or DNA sequences to enhance selectivity and performance for specific analytes.

Fabricated using a screen-printing technique, SPEs consist of conductive ink printed onto substrates through a woven mesh stencil, followed by thermal curing to solidify the layers. This method enables scalable and reproducible production while allowing for customiz in electrode design. However, challenges such as high-temper curing requirements and stencil fabrication remain constra Despite these limitations, SPEs have proven invaluable in environmen monitoring, biomedical diagnostics, and food safet g. The combination of portability, ease of use, and cost-e iven ensures their continued expansion in electrochemical s g, driv innovation in biosensing and point-of-care applications

4.4. Other electrodes

Platinum, gold, and silver electrodes ar dely used in electrochemical applications due to t excellent co ivity, stability, and resistance to corrosion. Platinum electrodes offer high durability and chemical inertness, making them ideal for long-term and high-temperature applications [143]. Gold electrodes provide superior conductivity and biocompatibility, and are often used in biosensors and nanotechnology [144]. Silver electrodes, particularly in the form of silver/silver chloride (Ag/AgCl), serve as reliable reference electrodes due to their stable potential. Additionally, modified electrodes, such as those coated with polymers, nanoparticles, or biomolecules, are tailored for specific analytical applications, enhancing sensitivity, selectivity, and catalytic properties in fields like environmental monitoring, medical diagnostics, and energy storage [145].

Table 1 summarizes the composition, key properties, advantages, limitations, and primary applications of diff ctrodes.

5. Modifications of GCE electrodes

5.1. Importance of GCE modifica

Unlike previous review at have adly c d electrode modifications for sensing app ons s review focuses specifically on GCE-based systems d o th espread , high conductivity, and modifiability. W ghlight e ing ds in GCE modification, including hyb composite ches, electrochemically activated syste and role in en ncing selectivity, stability, and sensitivity to environm pollutants [146]. The goal is to provide a focus o-date persp on recent GCE-specific advancements th e critical for pharmaceutical detection in water environments. M fying GCEs is pivotal advancement in electrochemistry, particula or detecting armaceutical pollutants in environmental samples. Un fied GCE ten lack the sensitivity and selectivity needed for precis making modification essential to enhance their performance [147].

importance of GCE modification lies in several key improve- nts offers. First, it enhances sensitivity by increasing the electrode's surface area, which is critical for detecting trace amounts of pollutants. For instance, a study showed that coating a GCE with multi-walled carbon nanotubes (MWCNTs) significantly improved the detection of pheniramine, an antihistamine drug, achieving a detection limit of 58.31 μg/mL [148]. Similarly, integrating nanomaterials like graphene oxide has been effective for paracetamol detection, reaching a detection limit as low as 4.9×10^{-8} M [149]. These modifications enable better adsorption and reaction of target molecules, ensuring the electrode can detect even minute quantities of pharmaceutical pollutants in water

Table 1
Comparison of different elec s.

Electrode Type	C ition	y Properties	Advantages	Limitations	Primary Applications
CPE	Mixtur carbon powe (e.g bon bes) a der (e. TILs)	Broad potential window, high signal-to-noise ratio, low ohmic resistance, hydrophobic	Easy modification, high sensitivity, renewable surface, chemically inert, environmentally friendly	Unstable in non-aqueous solutions, aging due to binder volatility, and heterogeneous surface	Sensing, environmental analysis, biosensors
GCE	lass-like carbon fro olic resins	High conductivity, chemical stability, wide potential window, and low residual current	Excellent for voltammetry, stable in various media, smooth surface after annealing	Sensitive to surface conditions, requires careful preparation, prone to oxidation in air	Voltammetry, electrochemical studies
SPE	Conductive inks on plastic/ ceramic substrates	High sensitivity, quick response, customizable design	Cost-effective, disposable, portable, mass-producible, adaptable for modifications	High-temperature curing needs and stencil fabrication challenges	Biosensing, environmental monitoring, diagnostics
Platinum Electrodes	Pure platinum metal	High conductivity, chemical inertness, durability	Long-term stability, suitable for high-temperature applications	Expensive, limited modification flexibility	Long-term studies, high-temperature applications
Gold Electrodes	Pure gold metal	Superior conductivity, biocompatibility	Excellent for biosensors, compatible with nanotechnology	High cost, softer material may limit durability	Biosensors, nanotechnology
Silver Electrodes (Ag/AgCl)	Silver coated with silver chloride	Stable potential, good conductivity	Reliable as reference electrodes, cost-effective compared to Pt/Au	Limited use beyond reference roles, less durable in harsh conditions	Reference electrodes in electrochemical cells

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[150].

Second, modification improves selectivity, which is vital in complex environmental samples containing multiple compounds that can interfere with detection. By introducing specific functional groups or recognition elements, modified GCEs can target particular pharmaceuticals while minimizing interference. A 2022 study demonstrated this with a pre-activated GCE combined with a silica nanochannel film, achieving high selectivity for p-nitrophenol with a detection limit of 0.02 μM [151]. Another example involves a GCE modified with MWCNTs and cyclodextrins, which selectively detected ibuprofen in water with a detection limit of $1.25 \times 10^{-} \times^{5}$ M, showcasing its applicability in environmental monitoring [152]. These advancements ensure that the electrode focuses on the pollutant of interest, enhancing accuracy [153].

Third, modified GCEs offer increased stability, preventing issues like electrode fouling that can occur in complex matrices. This stability ensures consistent performance over multiple measurements, a practical necessity for real-world applications. Electrochemical activation, for example, has been shown to enhance stability and catalytic properties, as seen in a study detecting 4-nitrophenol and dopamine in real samples [154]. Additionally, a GCE modified with cationic surfactants was used to detect roxarsone, an arsenic-based drug, demonstrating stable performance in environmental samples [155]. This reliability makes modified GCEs a dependable tool for ongoing water quality assessments [156].

Various modification techniques contribute to these benefits. Electrochemical activation involves cycling the electrode potential to alter its surface chemistry, boosting catalytic properties and sensitivity. Chemical coatings, such as polymers, metal oxides, or carbon-based nanomaterials like MWCNTs, tailor the surface for specific analytes. Nanomaterial integration, including graphene oxide or carbon n[illegible] tubes, enhances surface area and electron transfer kinetics. Compa[illegible] unmodified GCEs, which may exhibit high overpotentials and pa[illegible] ation, modified versions consistently show lower detection limits a[illegible] higher current responses, confirmed through methods [illegible] differenti[illegible] pulse voltammetry and cyclic voltammetry [157].

Beyond environmental monitoring, an unexp[illegible]d aspe[illegible] of modified GCEs is their versatility across medical and [illegible]trial [illegible] highlighting their broad utility. However, [illegible]lleng[illegible]ain, such [illegible] ensuring reproducibility and managing t[illegible]omplexity [illegible]ost of some modifications [158]. Cost-effective m[illegible] like electro[illegible]ical activation, offer solutions that balance [illegible]rfor[illegible]e and practic[illegible]ility. Ultimately, GCE modification is in[illegible]pensable fo[illegible]sitive, selective, and stable detection of pharmace[illegible]l pollutants, ma[illegible]it a cornerstone of environmental protection[illegible]rts [15[illegible]

5.2. Modification methods

As mentio[illegible] ele[illegible]de surfa[illegible]ification is essential to enhance the perform[illegible]e of ca[illegible]n-based el[illegible]rodes, particularly GCE. However, GCEs suff[illegible]om [illegible]s surface contamination, high overpotentials, an[illegible]nted redo[illegible] reactions, which can reduce sensitivity and reproducibi[illegible] To address these limitations, various modification techniques are em[illegible]d, including chemical functionalization, electrochemical treatments, and the application of nanomaterials or polymer coatings [160].

5.2.1. Chemical methods

Chemical modification of GCEs is a critical approach to tailoring their surface properties for specific electrochemical applications [161]. These modifications can be achieved through chemisorption, covalent bonding, or polymer film coatings, with or without the involvement of catalysts. Chemisorption relies on the adsorption of molecules onto the electrode surface through strong chemical interactions, which can range from hydrogen bonding to partial covalent character. The success of chemisorption depends mainly on the hydrophobic or hydrophilic compatibility between the GCE and the modifying agent, influencing not only the strength of adhesion but also the uniformity and durability of the resulting film. This method is often favored for its simplicity and ease of implementation, although the strength of attachment can vary depending on environmental conditions [162,163].

Covalent modification, in contrast, offers a more stable and permanent means of altering the GCE surface. Functional groups on the carbon surface—such as hydroxyl, carboxyl, or amine groups—can react with modifying reagents to form strong, irreversible covalent bonds [164]. This leads to the creation of monomolecular layers that are highly resistant to desorption, even under electrochemical stress. [165]. Typical reaction conditions include mild heating (25–60 °C), use of coupling agents such as EDC/NHS for amid[illegible] formation, and reaction times ranging from 1 to 24 h under [illegible]ert a[illegible]here to prevent oxidation. To enhance this stability f[illegible]er, bifunct[illegible] crosslinking agents such as glutaraldehyde, ethylene[illegible]ol diglycid[illegible]ther, or isothiocyanates are often employ[illegible] These[illegible]pounds[illegible]rm covalent bridges between the electrod[illegible]rface and fu[illegible]n[illegible]olecules, reinforcing the structural inte[illegible] of the[illegible]dified [illegible] This is especially valuable in biosensor dev[illegible]men[illegible]here enzyme immobilization requires both strong at[illegible]men[illegible]etention[illegible]ological activity [166]. Polymer film coati[illegible] represent[illegible]hly [illegible]atile modification strategy. Depending on[illegible]sired functi[illegible], one may choose electron-conductive [illegible]yme[illegible]g., polyaniline, polypyrrole) to facilitate charge transfer or inst[illegible]g polymers to act as selective barriers or matri[illegible]urther functi[illegible]zation. As shown in Fig. 5, polymer films ad[illegible] through a combinatio[illegible] of chemisorption and precipitation. Dip co[illegible]ng—immersi[illegible] the electrode in a polymer suspension—enables fa[illegible] film forma[illegible], with thickness and morphology tuned via suspen[illegible] concentr[illegible]n, viscosity, and immersion time. Reaction conditions [illegible]volve room temperature to 50 °C immersion, suspension concentrations between 0.1 and 10 wt%, and immersion [illegible]rom seconds to several minutes, followed by drying under [illegible]bie[illegible] or controlled humidity conditions. To overcome the non-uniformities that can arise from dip coating, spin coating can be employed to generate smoother, more consistent films by using centrifugal forces to spread the polymer solution evenly [167]. The performance of polymer-coated GCEs can be dramatically enhanced by incorporating nanomaterials such as carbon nanotubes, graphene, or metal nanoparticles into the polymer matrix (Fig. 4). These additives boost the effective surface area, improve electron transfer kinetics, and introduce catalytic sites, thereby elevating sensitivity, selectivity, and overall electrochemical activity. Through careful selection of modifiers, crosslinkers, and deposition methods, researchers can precisely engineer GCE surfaces to achieve long-term stability and high performance in applications ranging from biosensing and environmental monitoring to electrocatalysis and energy storage [168].

A representative example of covalent chemical modification of GCEs involves the electrochemical grafting of organic molecules via in situ-generated diazonium salts. In this approach, various aryl diazonium cations—produced by reacting aromatic amines with $NaNO_2$ and HCl—were electrochemically reduced onto the GCE surface using either cyclic voltammetry or potentiostatic methods in an acidic aqueous medium. Specifically, diazonium salts were generated at 0–5 °C in 0.5 M HCl, and the electrochemical grafting was performed by cyclic voltammetry scanning between + 0.6 V and − 0.2 V vs. Ag/AgCl at scan rates of 50–100 mV/s for 3–10 cycles, or potentiostatic reduction at potentials around − 0.3 to − 0.5 V for 30–300 s [169]. Functional groups such as $–OPO_3H_2$, $–SO_3H$, –COOH, –OH, $–NO_2$, and –Br were introduced through corresponding diazonium precursors, forming compact and stable aryl-modified layers covalently attached to the electrode. Electrochemical techniques including cyclic voltammetry and electrochemical impedance spectroscopy (EIS) confirmed the successful modification and blocking properties of the resulting films, while linear sweep voltammetry (LSV) demonstrated distinct electrochemical responses toward paracetamol (PCT) depending on the terminal group

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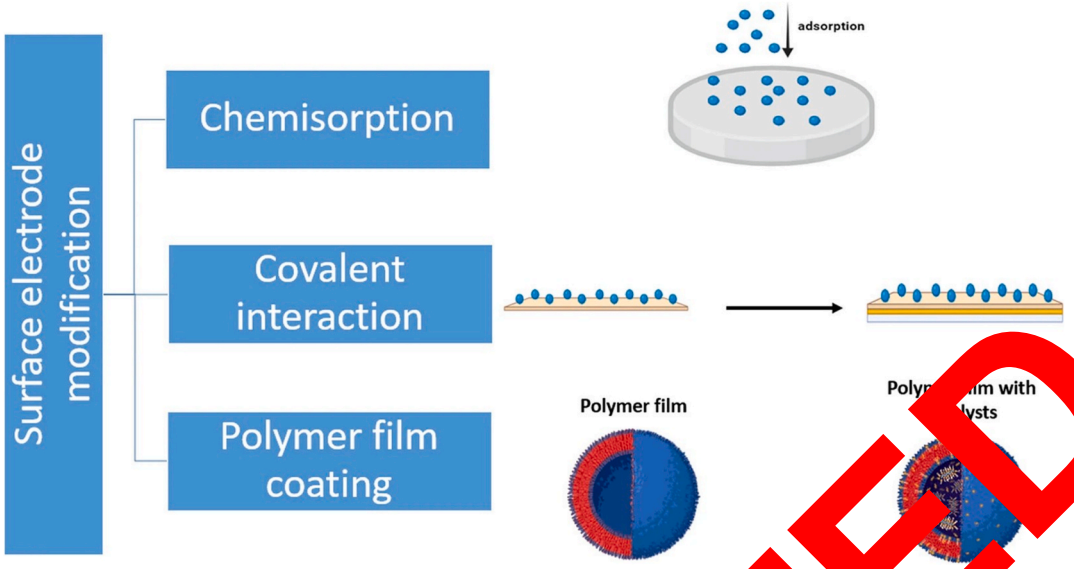


Fig. 4. Schematic of chemical methods for modification of GCE [167].

[170]. Specifically, electrodes modified with electroactive terminal groups exhibited oxidation peaks for PCT similar to those on bare GCEs, whereas inert group-modified electrodes behaved like microelectrodes, showing no oxidation signal. These differences were attributed to variations in terminal group electronegativity, pKa, and their hydrogen bonding ability with PCT. This example underscores the versatility of diazonium-based grafting in tuning GCE surface properties for specific sensing applications [171].

5.2.2. *Physical methods*

Physical methods for modifying GCEs involve the non-covalent attachment of functional or active agents to the electrode surface. These interactions are typically governed by weak intermolecular forces, including electrostatic interactions, hydrogen bonding, π–π stacking interactions, and Van der Waals forces (Fig. 5). Because these interactions do not involve the formation of covalent bonds, physical modifications are generally simpler, reversible, and often do not alter the intrinsic electronic structure of the GCE. As such, these methods are frequently utilized in diverse fields such as solar energy conversion, molecular electronics, biosensing, and electroanalytical chemistry [172].

A common approach within physical modification is encapsulation, where the active modifying agent is physically entrapped within a host matrix or membrane, such as a sol–gel, Nafion film, or polymeric network. This method not only protects the modifying agent from environmental degradation but also provides a confined environment that may enhance its activity and selectivity. Encapsulation is particularly useful in biosensor development, where enzymes or proteins are incorporated into biocompatible matrices to maintain their functionality and improve signal stability. Another widely used strategy is the application of polymer coatings, which involves coating the electrode surface with thin films of conductive or insulating polymers. These

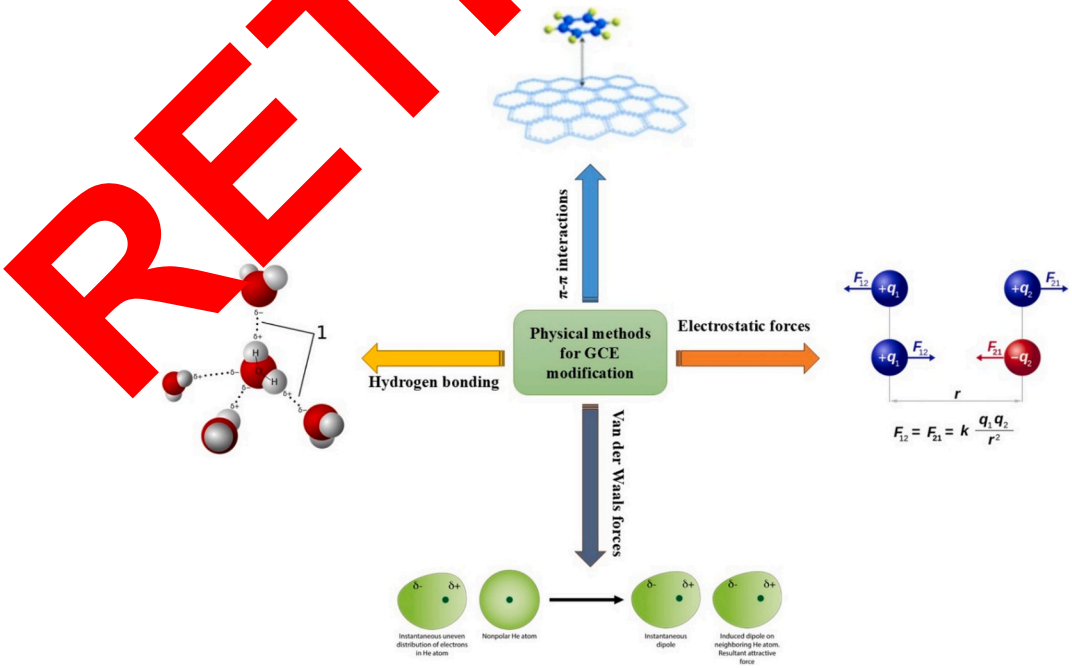


Fig. 5. Schematic of chemical methods for modification of GCE.

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coatings serve multiple purposes: they can act as selective barriers, enhance charge transfer, immobilize other active species, or provide specific functional groups for further chemical modification. Polymers such as polyaniline, polypyrrole, and polythiophene derivatives are commonly employed due to their electrical conductivity and ease of film formation [173]. Adsorption techniques are also prevalent, wherein surface-active substances are allowed to adsorb onto the GCE surface through physical attraction. These substances may include dyes, metal complexes, surfactants, or nanoparticles. This approach offers simplicity and minimal processing, and the adsorbed layer can impart specific chemical or electrochemical properties to the electrode surface. Despite the many advantages of physical methods—such as simplicity, cost-effectiveness, and minimal requirement for chemical reagents—these techniques are not without drawbacks. The weak forces that anchor the modifying agents to the electrode surface often result in limited mechanical stability, making the modified surfaces susceptible to desorption, especially under harsh electrochemical conditions. Additionally, anisotropic distribution of the modifying phase can lead to inhomogeneous surface coverage, which negatively affects reproducibility and performance consistency across different electrode batches. Moreover, the lack of strong bonding may impair long-term operational stability, a critical factor for practical applications such as continuous monitoring or in vivo sensing [174].

In conclusion, physical methods offer a flexible and straightforward approach to GCE modification, particularly for applications that benefit from reversible or non-permanent functionalization. However, for applications demanding high durability and reproducibility, these methods may require further optimization or combination with more robust chemical modification techniques.

5.2.3. Electrochemical methods

Electrochemical methods for modifying GCEs primarily involv deposition of metal nanostructures or the formation of polymer film the electrode surface. These processes are typically controlled us potentiostatic or potentiodynamic techniques, which w preci tuning of the material properties at the electrode int ce. specific electrochemical modifications enhance the perfo nce of GCE in various applications such as sensors, batteries, cell treatment systems by improving electron tra r, ca activity, a surface characteristics [175].

In potentiostatic deposition, a co otential is a d to the electrode surface, which induces th educ of metal ions present in the electrolyte solution. The pr ss typically v ves the formation of metal nanostructures at speci efect sites on the . The reduction of metal ions occurs prima at thes defect sites d e to the localized increase in electric field re nu tion and growth of metal particles or clusters begin. The de time, el olyte concentration, and applied voltage a ical fa in de ining the final morphology of the deposit met uch as s hape, and uniformity [176]. A longer dep on tim often lead o thicker deposits, while higher electrolyte centr sult in faster metal ion reduction rates. The potentia in potentiostatic deposition ensures that the metal nanostructures r d are highly uniform, and by adjusting the voltage and deposition time searchers can tailor the nanostructures to specific applications, achieving precise control over surface characteristics like roughness and porosity. Typical reaction conditions for potentiostatic deposition include applied potentials ranging from − 0.2 V to − 1.0 V vs. Ag/AgCl, electrolyte metal ion concentrations between 1 and 10 mM, deposition times from 30 s up to 10 min, and temperatures maintained at room temperature (20–25 °C) [177].

Potentiodynamic deposition, often performed using CV or DPV, involves the scanning of the potential over time, which allows the control of metal structure formation based on the changes in the electrode's electrochemical environment. In this method, the potential is varied continuously, and as the potential sweeps through different ranges, metal ions undergo reduction and deposition at the electrode surface. This dynamic process enables the creation of metal films with specific morphologies and thicknesses by carefully controlling the rate of potential change and the number of cycles [178]. The potentiodynamic deposition technique is particularly advantageous for forming metal nanostructures with defined shapes and sizes, as the constant scanning of the potential leads to the gradual and controlled growth of the metal structures. CV or DPV can also be used to monitor the electrochemical behavior of the system during the deposition process, providing insights into the electrodeposition mechanisms and helping researchers optimize the experimental conditions for desired properties. Electrochemical parameters typically include potential scan rates from 10 to 100 mV/s, potential windows between − 0.5 V to + 0.5 V vs. Ag/AgCl, metal ion concentrations of 1 to 5 mM, and 10 to 50 ion cycles, all performed at ambient temperature [179].

In electropolymerization, a poten is applied polymerize a monomer, forming an adherent polyme on the G surface. The properties of the polymer film, as thi s and c uctivity, are influenced by factors like sol choice, mo r centration, and applied potential. This pr is ide r creat ms that improve the GCE's electrochemica bilit d performance in various applications. This method olve ly cont g the potential, current density, electroly ncentrati d d sition time to create metal nanostructures g from 1 to on the GCE. The formation of these nanos tures eases the su ace area and enhances catalytic and electrochemical pr ies, making them ideal for applications in senso cells, and en storage devices [180]. Electrochemical mo cations of GCEs signi antly enhance their performance. Metal na structures an polymer films increase surface area, promote effici electron tra r, and improve catalytic properties, which lead to bet rforman n electrochemical applications. These modifications make sensitive and selective, improving the efficiency of biosensors, energy storage devices, and environmental monitoring systems. Typical electropolymerization conditions involve applied potentials ranging from + 0.7 V to + 1.2 V vs. Ag/AgCl, monomer concentrations between 5 and 50 mM, scan rates of 20 to 100 mV/s, and polymerization times of 5 to 30 min under room temperature conditions [181].

5.2.4. Soft plasma polymerization (SPP)

SPP is an innovative and emerging surface modification technique that utilizes low-temperature plasma to deposit ultrathin polymer films directly onto substrates such as GCEs. Unlike conventional wet chemical or electrochemical modification methods, SPP is a solvent-free, dry process that enables high precision and reproducibility. In this method, gaseous monomers are introduced into a plasma discharge region, where they become activated and polymerize upon contact with the electrode surface. The resulting film is conformal, adherent, and rich in functional groups, providing significant advantages for electrochemical applications [182]. A key benefit of SPP for GCE modification is its precise control over surface properties, such as film thickness, uniformity, and chemical functionality, through simple adjustment of process parameters like plasma power and deposition time. Moreover, SPP is compatible with thermally or chemically sensitive materials, including biological macromolecules, which makes it ideal for biosensing applications. The resulting surfaces often exhibit enhanced stability, improved resistance to fouling, and better long-term performance under environmental conditions. These attributes make SPP a promising tool for fabricating GCEs with superior sensitivity, selectivity, and robustness [183].

The utility of SPP is clearly illustrated in the fabrication of laccase-based biosensors. As shown in Fig. 6, the laccase bio-recognition layer was deposited onto the GCE surface using a corona discharge jet in the SPP process. SEM imaging reveals a distinct correlation between deposition time and surface morphology. After 10 s of plasma exposure (Fig. 6a), extensive but sparsely distributed laccase clusters were formed. Increasing the deposition time to 15 and 30 s (Fig. 6b and c) resulted in more uniform and denser distribution of laccase clusters

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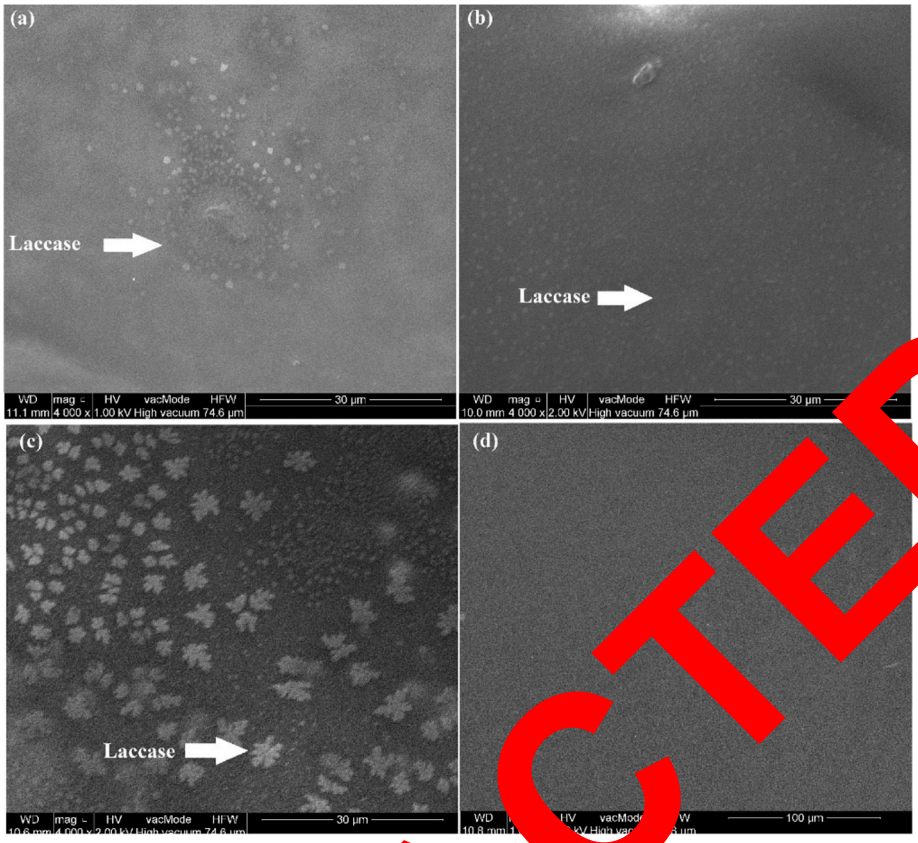


Fig. 6. Scanning electron microscope (SEM) images of laccase layers de (a) 10 s, (b) (c) 30 s by corona SPP deposition process and (d) unmodified GCE [184].

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across the GCE surface. Notably, the 30-second SPP pro produc bio-recognition layers with complex cluster stru dicating enhanced surface coverage and likely improved zymati ctivity. These observations confirm that SPP allows for tunin lec- trode surface at the micro- and nanoscale, hich mizing biosensor performance [184].

In conclusion, SPP offers several cr dvantages o raditional modification techniques: it is clean itr le, and comp ible with delicate biomolecules, and it is apable of pr ing highly functional and stable surfaces. Its abilit ailor the elect interface precisely makes it particularly att ve for next-generat electrochemical sensors and biosensors integr of such advanced modification strategies into GCE platfo ly enha es analytical performance but also broadens the scop environ tal and biomedical moni- toring applicat

5.2.5. La blation

Laser ab d surface modification technique that has gained sign t attention for enhancing the performance of GCEs in electrochemica lications. This method involves directing a high-energy laser beam o the electrode surface to remove material in a highly localized manner selectively. The process not only alters the physical topography of the GCE—typically increasing its surface roughness and porosity—but also induces chemical changes, such as the formation of functional groups or defect sites that can promote electrocatalytic activity. One of the main advantages of laser ablation is its precision and controllability; by adjusting parameters such as laser power, pulse duration, and exposure time, researchers can fine-tune the extent and nature of the surface modification [185]. Unlike chemical or electrochemical treatments that may involve hazardous reagents, laser ablation is a clean, reagent-free, and rapid process, which makes it attractive for sustainable and scalable sensor fabrication. Moreover, it allows for patterning or structuring the electrode surface with minimal thermal damage, preserving the integrity of the bulk material while significantly enhancing its electrochemical properties. As a result, laser ablation is increasingly being adopted as a straightforward and effective strategy for the development of high-performance electrochemical sensors [186].

In the context of electrochemical sensing, laser-modified GCEs (LGCEs) offer several key advantages. For example, in the development of a novel electrochemical sensor for detecting ofloxacin (OFL)—a widely used fluoroquinolone antibiotic in environmental water samples—laser ablation was used to modify the GCE surface structurally. The process was optimized at a laser power of 1.8 W and a duration of 40 s, under a slightly acidic pH of 5.5. Post-modification, the LGCE was thoroughly characterized using SEM and atomic force microscopy (AFM), which revealed an increase in surface roughness. This increased roughness is crucial as it enhances the effective surface area of the electrode, allowing for more active sites for electron transfer. Additionally, X-ray diffraction (XRD) and X-ray photoelectron spectroscopy (XPS) confirmed the presence of new functional groups—such as oxygen-containing species—on the electrode surface [187]. These functional groups contribute to improved electrochemical reactivity and promote stronger interactions with OFL molecules. EIS further demonstrated a decrease in charge transfer resistance, indicating enhanced conductivity and faster electron kinetics due to laser treatment. Functionally, the modified electrode enabled sensitive detection of OFL via DPV at a working potential of + 0.91 V vs. Ag/AgCl. The response was linear over a wide range (0.25–200 μM) with an impressively low detection limit of 75 nM, without the need to deoxygenate the samples. This not only simplifies the analytical process but also makes the sensor more suitable for field applications. The LGCE sensor was successfully applied to real samples, including spiked groundwater, tap water, and wastewater, achieving high recovery rates (94–108 %) and excellent

reproducibility (RSD < 4.8 %) [188].

In summary, laser ablation provides a powerful route for modifying GCEs, introducing desirable topographical and chemical surface features that significantly boost sensor performance. This makes it a promising approach for developing high-performance electrochemical sensors for environmental monitoring and other applications.

5.2.6. Emerging material combinations for GCE modification

Recent advances in material science have led to the development of novel material combinations that significantly enhance the performance of GCEs by synergistically integrating the desirable properties of multiple components [189,190]. These emerging hybrid systems typically involve the integration of nanomaterials such as MOFs, covalent organic frameworks (COFs), MXenes, 2D layered materials (e.g., black phosphorus, transition metal dichalcogenides), and conductive polymers, often in conjunction with metal nanoparticles, CNTs, or graphene derivatives. The primary goal of such combinations is to simultaneously improve conductivity, surface area, electrocatalytic activity, and chemical functionality [191].

Hybrid Nanocomposites: One of the most widely explored strategies is the fabrication of nanocomposites that incorporate carbon-based nanostructures with metal oxides or nanoparticles [192,193]. For instance, rGO–Fe_3O_4, CNT–ZnO, or graphene-Au nanocomposites have shown enhanced charge transfer kinetics and improved active surface area due to their hierarchical architectures. These composites leverage the high conductivity and mechanical stability of carbon allotropes along with the redox activity or catalytic properties of metal components. 2D Material Heterostructures: Another emerging class involves vertical or lateral heterostructures formed by stacking different 2D materials, such as MoS_2/graphene or Ti_3C_2-MXene/g-C_3N_4. These structures provide abundant edge sites and interfacial charge transfer pathways, significantly enhancing the sensitivity and selectivity of based sensors. MXenes, in particular, offer a tunable surface chem and exceptional metallic conductivity, making them highly attractive electrochemical applications. Conductive Polymer–Nano le Blend Recent studies have also focused on blending conduc po rs (e.g., PEDOT:PSS, polyaniline) with noble metal nano cles (e Pt, Au, Ag) to form multifunctional coatings [194]. Th nate the processability and film-forming ability olyn n the exce lent catalytic activity of metal NPs. T esulting osite layers exhibit improved signal amplificati tifouling pr ies, and enhanced biocompatibility, which a esse or biosensing platforms [192]. MOF/COF-Based System Metal–organi meworks and covalent organic frameworks ha merged as highly ous and functionalizable materials that n be d either directly as electrode modifiers or as scaffold ost o active species. For instance, GCEs modified with MOF-derive materi r COF–AuNP composites have demonstrat prove troca ic properties, high surface area for immo ing molecu controllable pore structures for molecular cognitio Synergisti unctionalization: Some studies explore ti pon multi-component systems, such as graphene–polyp e_3O_4 o ene–AuNP–aptamer constructs, where each componen tributes distinct electrochemical functionalities. These systems are ored for specific target analytes (e.g., heavy metals, neurotransmi rs, or antibiotics) and show excellent analytical performance in terms of low detection limits, wide linear ranges, and high reproducibility. A recent work utilized a GCE modified with a nanocomposite of black phosphorus quantum dots (BPQDs), polyaniline (PANI), and AgNPs to detect dopamine. The BPQDs provided high electron mobility and abundant active sites, while PANI enhanced film stability and AgNPs served as electrocatalysts. The synergistic effect of the materials led to a detection limit of 20 nM with excellent selectivity even in the presence of ascorbic acid and uric acid—commonly interfering species [195].

In general, the rational design of hybrid and hierarchical material systems for GCE modification opens up new opportunities for tailoring electrode surfaces with superior physicochemical properties. These emerging material combinations hold great promise for next-generation electrochemical devices, especially in fields requiring high-performance sensing, such as point-of-care diagnostics, environmental surveillance, and food safety monitoring [196].

Traditional methods—such as chemical, physical, and electrochemical modifications—have been widely employed due to their simplicity, accessibility, and well-established protocols. Chemical methods provide strong and stable covalent functionalization but often involve complex procedures and potential degradation of biomolecules. Physical methods are favored for their ease of use and reversibility but suffer from limited stability. Electrochemical methods offer precise control over deposition and surface archi though they are restricted by electrolyte conditions and i ramen requirements. In contrast, advanced approaches such SPP, Laser tion, and the integration of emerging material combi ns introdu ew possibilities in GCE engineering. These n r techni s enable h-resolution surface control, improved b mpatibility, functional interfaces but may involve h er cost ynthesis plexity, or reproducibility issues. A side ide parison of these approaches is presented in Table rovi sight in eir underlying mechanisms, practical a ntages, li ions d application scopes. This comparative f k underscor evolution of GCE functionalization strate s fro ventional to cutting-edge technologies.

6. T nanostructu or GCE modification

ifferent types f nanostructures, as shown in Fig. 7, are widely used fo E modifica to enhance its electrochemical performance. Metal nan ticles (e Au, Ag, Pt, Pd) exhibit excellent conductivity, catalytic and biocompatibility, making them ideal for biosensing and electrocatalysis [197]. Carbon-based nanostructures, lag graphene, CNTs, and carbon dots, offer a high surface area, per electrical conductivity, and excellent mechanical properties, facilitating electron transfer. MOFs are highly porous materials with tunable structures, providing large surface areas and selective adsorption capabilities, which enhance sensitivity in electrochemical applications. Hybrid nanostructures and composites combine different nanomaterials, leveraging the synergies of metals, carbon-based materials, and MOFs to improve stability, conductivity, and catalytic efficiency, making them highly versatile for sensor development and energy storage applications.

6.1. Metal (Au, Ag, Pt, Pd, etc.) and metal oxide nanoparticles

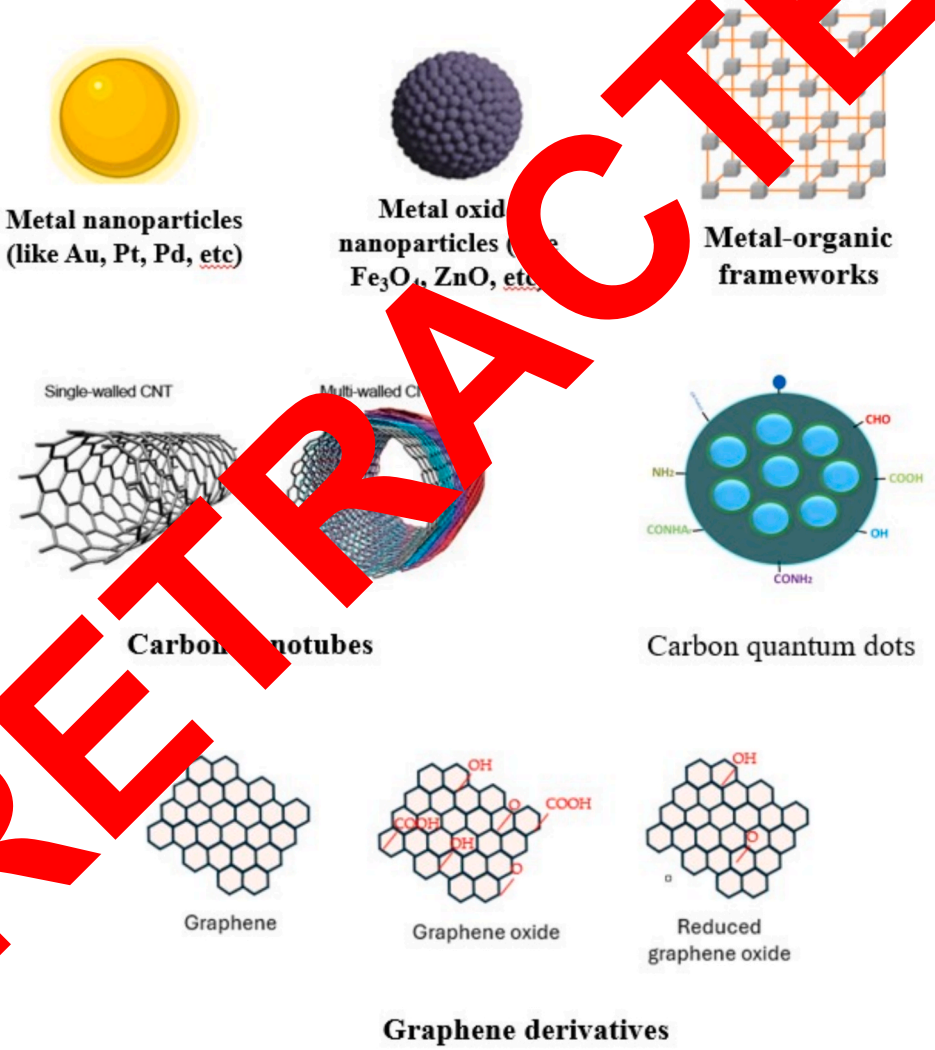
Modification of the GCE using metal and metal oxide nanoparticles has become a widely explored strategy in pharmaceutical sensing, particularly for the detection of biologically active compounds [198]. Metal nanoparticles, such as gold (Au), silver (Ag), platinum (Pt), and palladium (Pd), possess unique electronic, optical, and catalytic properties due to their nanoscale dimensions and high surface-to-volume ratio [199]. These characteristics make them ideal candidates for enhancing electrochemical sensor performance [200].

Among these, AuNPs have shown exceptional promise due to their biocompatibility and excellent conductivity. For instance, Li and Lin developed an electrochemical sensor by electrodepositing gold nanoclusters onto a polypyrrole (PPyox) film-modified GCE. The negatively charged PPyox film facilitated the accumulation of analytes via π-π interactions, and the catalytic oxidation of target molecules was enhanced at the gold nanocluster sites. This resulted in distinct and well-separated voltammetric peaks for compounds such as DA and ascorbic acid (AA) [201]. Similarly, Wei et al. employed a self-assembly approach using L-cysteine-modified GCE and AuNPs, allowing for selective detection even in the presence of multiple interfering substances such as epinephrine and folic acid [202]. Choudhary et al. further advanced this concept by creating a gold nanoparticle–amino acid composite, demonstrating

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Table 2
Comparison of different methods for modification of GCEs.

Method	Mechanism	Advantages	Disadvantages	Typical Applications
Chemical Methods	Covalent bonding between GCE surface groups and functional molecules	High stability, irreversible modification, suitable for enzyme immobilization	Multi-step processes, potential biomolecule activity loss	Biosensors, catalytic surfaces, enzyme-based sensors
Physical Methods	Physisorption via van der Waals, electrostatic, or hydrogen bonding forces	Simple, fast, low-cost, no chemical reagents	Low stability, desorption under varying pH or temperature	Preliminary sensor development, temporary coatings
Electrochemical Methods	Electrochemical deposition of polymers, metals, or nanoparticles	Precise control over film thickness, clean process, reproducible	Requires electrochemical setup, limited to specific electrolytes	Electrochemical biosensors, nanostructured electrodes, electrocatalysis
Soft Plasma Polymerization (SPP)	Plasma-assisted vapor-phase polymerization on the electrode surface	Solvent-free, uniform coatings, preserves biomolecule function	High equipment cost, parameter-sensitive process	Biocompatible sensors, high-perform[illegible] biosensors
Laser Ablation Method	Surface structuring by laser-induced material removal	High precision, localized surface modification, patternable	Risk of surface damage, requires careful optimization	[illegible]structure[illegible]rodes, high-[illegible]face-area sens[illegible]
Emerging Material Combinations	Hybrid materials (e.g., MOFs, COFs, MXenes, nanocomposites) integrated onto GCE	Synergistic effects, high surface area, multifunctional properties	Complex synthesis, scalability issues, reproducibility challenges	[illegible]nced biosensi[illegible]ultiplexed de[illegible], next-gen[illegible]trochemical interf[illegible]



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Fig. 7. Numerous materials are frequently employed in GCE surface modification.

excellent catalytic efficiency for pharmaceutical targets [203].

To overcome the limitations of molecularly imprinted polymers (MIPs), such as poor conductivity, researchers like C. Xue et al. integrated AuNPs into MIPs layered on reduced graphene oxide (rGO) and polyaniline (PANI) nanocomposites. This modification enhanced the electrocatalytic activity and sensitivity of the sensor, achieving a low detection limit. Among the various combinations studied, AuNPs in conjunction with polypyrrole film produced the best results in terms of sensitivity and detection limits. Despite their high performance, the cost of gold remains a limitation for widespread use [204].

In addition to gold, silver nanoparticles have been explored due to their high conductivity and biocompatibility. Sadanandhan et al. developed a composite electrode consisting of poly(3,4-ethylenedioxythiophene) (PEDOT), rGO, and AgNPs. This hybrid electrode exhibited an impressive detection limit for pharmaceutical analytes, along with excellent stability, reproducibility, and a broad linear detection range. The PEDOT nanotubes effectively acted as conductive bridges, enhancing electron transfer between the RGO sheets and AgNPs [205].

Metal oxide nanoparticles have also attracted significant interest due to their unique surface properties and catalytic behavior. Titanium dioxide (TiO_2) nanoparticles, known for their high surface area and biocompatibility, were combined with MWCNTs and ionic liquid by Mazloum-Ardakani and Khoshroo. This sensor, while easy to fabricate and applicable over a wide concentration range, had a relatively higher detection limit compared to Au-based sensors [206]. Ran et al. introduced magnetic nanoparticles and fabricated a composite film with poly (bromocresol green) and MWCNTs on a GCE. The conducting polymer poly (BCG) carried a negative charge, facilitating enhanced electrocatalytic activity for positively charged analytes [207].

Further, Anithaaa et al. modified GCE with irradiated tungsten trioxide (WO_3) nanoparticles. Exposure to a 100 kGy radiation dose altered the crystal structure and increased the surface area of the WO_3 nanoparticles, leading to improved electron transfer and high sensitivity. Although this method produced high-performing sensors, its non-eco-friendly synthesis posed challenges. In another notable study, Dinesh et al. employed cobalt oxide (Co_3O_4) nanoparticles in combination with rGO to modify GCE via a simple electrochemical approach. The resulting sensor displayed significantly enhanced peak current responses, indicating strong synergistic effects between Co_3O_4 and RGO [208].

Generally, the integration of metal and metal oxide nanopar with conductive polymers and carbon-based nanomaterials on GCE s significantly advanced the field of electrochemical sensing. Th modifications not only enhance the sensitivity and se ity of th sensors but also extend the linear detection ran d ce the detection limits for various pharmaceutical and iological elevant molecules. While metal nanoparticles, such as Au Ag, performance, their high cost and synthesis lleng their pra tical applications. Conversely, metal ox nanoparti when combined with materials like rGO or CNTs cost-effectiv efficient alternative, although they still face dles nthesis com exity and biocompatibility. These innov ions reflect ignificant trend in biosensor development, pus the boundaries recision and reliability in electrochemical ection.

6.2. Carbon-based nanostr graphene rbon nanotubes, carbon dots)

Modifi n of G with ca n-based nanostructures, such as CNTs, gra e, a s, has significantly advanced electrochemical sen pharmaceutical applications, particularly for neurotransmitters l rotonin (5-HT, ST) and dopamine (DA). Due to their nanoscale di ions (1–100 nm), these nanomaterials possess high surface-to-volum ratios and excellent electrical conductivity, enhancing electron transfer kinetics and sensitivity. MWCNTs, in particular, offer robust electrocatalytic activity and have been widely employed with various supporting materials, such as ionic liquids, chitosan, and Nafion, to improve electrode stability, dispersion in aqueous media, and signal resolution. For example, MWCNTs combined with dihydrogen phosphate or ionic liquids demonstrated well-separated oxidation peaks for ST and DA, high selectivity, and significantly lower detection limits compared to unmodified electrodes. Chitosan-functionalized CNTs also enhanced stability and diffusion rates, though excessive amounts could dampen peak currents [209].

Graphene and its derivatives, such as GO and rGO, have also emerged as powerful electrode modifiers due to their exceptional conductivity, mechanical strength, and large surface area. However, the hydrophobic nature of pristine graphene and its tendency to agglomerate necessitate functionalization or hybridization with polymers or nanoparticles to enhance solubility and stability. The electrochemical reduction of GO to rGO significantly improves conductivity, making it suitable for the electrochemical detection of serotonin with high sensitivity and a broad linear detection range. Furthermore, nitrogen-doped or porphyrin-functionalized GO composites offer additional active sites and improved electrocatalytic activity. These carbon nanomaterial-based modifications have outperformed traditional electrode systems, demonstrating high reproducibility, low background currents, and adaptability for biosensing platforms, positi em as ideal candidates for sensitive and selective pharmac al an [210].

6.3. MOFs

GCE modification using s has show nt promise in pharmaceutical applicati due t OFs' u e structural and chemical features. Their i por , large surface area, and tunable functional groups pr e an platform electrochemical sensing by enhancing the ber of ac ites ilable for molecular recognition. These s enable m ient signal transduction and sensitivity, ch a itical in pharmaceutical analysis [211]. For instance, Daud et al. de trated the utility of MOFs in developing an enzy amperometri sor for glucose monitoring, utilizing a Ni C–NH_2 framework. This structure allowed effective glucose di ion and ads tion via its multi-layered architecture and amino-fu onalized li s, highlighting the potential of MOF-modified GC r precise d long-term biomolecule detection in pharmaceutical c].

lymer-based composites

GCE modification using polymer-based composites plays a crucial role in pharmaceutical electrochemical sensing by enhancing electrode interface stability and mechanical strength. Polymers, both natural (e.g., chitosan, cellulose) and synthetic (e.g., EVOH, acrylic, conducting polymers), can directly participate in sensing or serve as immobilization matrices for specific receptors. These matrices support both non-covalent and covalent immobilization strategies, enabling the stable incorporation of biomolecules. Despite their low intrinsic conductivity, polymers are widely used due to their biocompatibility and functional versatility, with their conductivity limitations often mitigated by forming composites with conductive materials [213]. Notable applications include a shrink polymer-based sensor using PLL on a graphene-modified GCE for detecting PSA biomarkers, showing high selectivity and clinical diagnostic compatibility, and a CP precursor (TBA)-based thrombin sensor, where the polymerized matrix on GCE enabled precise thrombin detection via a sandwich-type immunoassay. These modifications highlight the promising integration of polymer-based composites in the development of reliable, sensitive, and clinically viable electrochemical sensors in pharmaceutical contexts [214].

7. Electrochemical sensing of pharmaceutical pollutants

Electrochemical detection of pharmaceuticals using GCE modified with carbon nanomaterials has shown significant promise due to their enhanced electrocatalytic properties [215]. Carbon-based materials like SWCNTs and rGO have been employed in electrochemical sensors to detect various pharmaceuticals, such as acetaminophen, methylglyoxal, and valacyclovir, with high sensitivity and low detection limits. In the case of acetaminophen detection, as shown in Fig. 8a, rGO sheets were deposited on GCE, exhibiting a characteristic rippled structure that improved the sensor's surface area. The electrochemical behavior of 50 μM acetaminophen on rGO/GCE, presented in Fig. 8b, demonstrated

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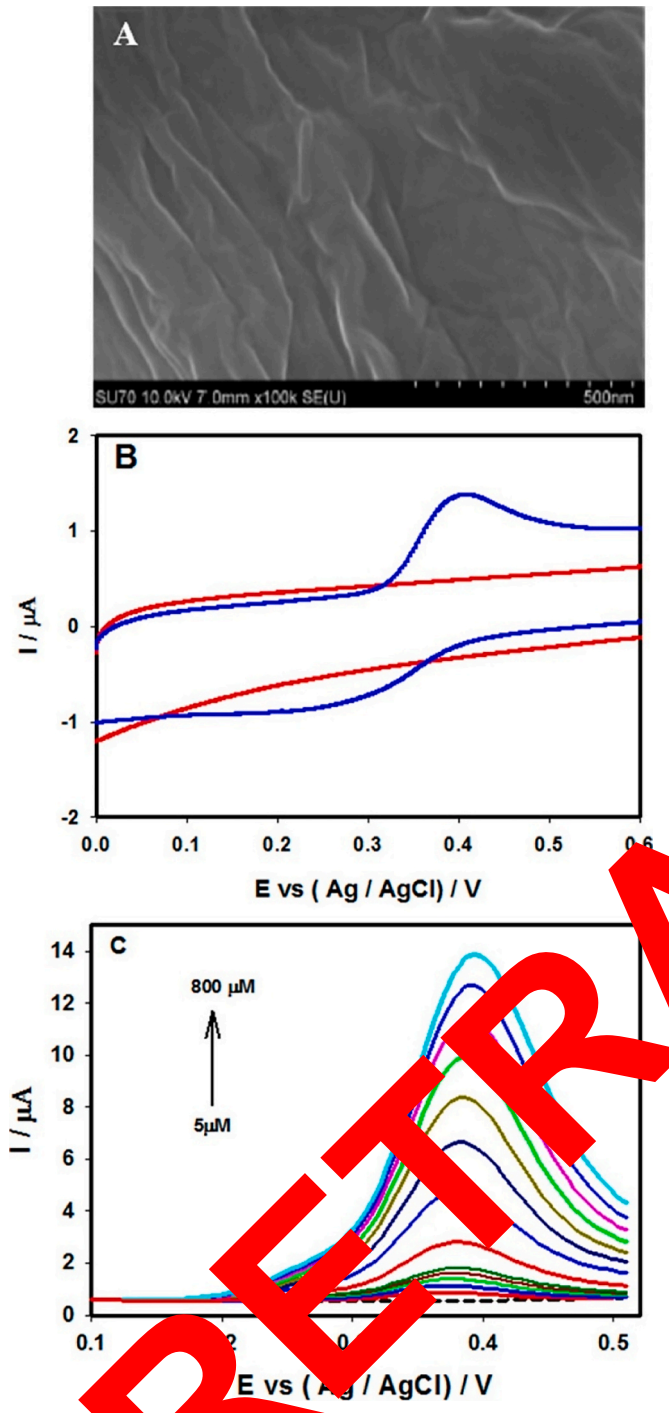


Fig. 8. Electr… rGO… the GCE surface is shown in (a) FE-SEM images; (b) CVs … rGO/GCE in 0.1 M PBS (pH 7.4) with and without 50 µM acetaminophen … solid line) at a scan rate of 20 mV/s; (c) DPVs of the rGO/GCE in 0.1 M PBS … 7.4) with varying concentrations of acetaminophen (5 µM to 800 µM) [216].

distinct redox peaks, indicating a reversible electrode process. Fig. 8c further illustrates that the current responses increased linearly with the concentration of acetaminophen, ranging from 5 µM to 800 µM, with a strong correlation coefficient ($R^2 = 0.996$). The LOD for acetaminophen using rGO/GCE was found to be 2.1 nM. This study highlights the superior performance of rGO-based electrochemical sensors, which achieve high sensitivity, low detection limits, and excellent linearity, making them valuable tools for the sensitive and rapid detection of pharmaceutical compounds [216].

Detection of acetaminophen, a widely used antipyretic and analgesic drug whose overdose poses serious health risks, can be effectively achieved using an electrochemical sensor based on a NiCoFe layered double hydroxide composite integrated with graphitic carbon nitride (NiCoFe LDH@g-CN). This sensing platform exploits the synergistic interactions between NiCoFe LDH and g-CN, which enhance electron transfer kinetics and provide abundant active sites through multivalent metal centers and nitrogen-containing groups, resulting in improved conductivity, catalytic activity, and analyte adsorption via π–π interactions and hydrogen bonding. Electrochemical studies, including CV and EIS, demonstrate that the NiCoFe LDH@g-CN-modified GCE exhibits a significantly lower charge transfer resistance of 418 Ω and a higher electron transfer rate constant (k_0) of 1.27 × … s^{-1} compared to unmodified or singly modified electrode… V a… reveals a low detection limit of 0.06 µM and high se… vity of 2.43… A μM^{-1} cm^{-2} for AP detection over a wide linear ra… of 0.166–1… .5 µM, confirming the sensor's excellent an… ical pe… nance. F… hermore, the sensor shows high selectivity… inst comm… te… ng substances, reproducibility with RSD a… d 2.3… %, stab… over 15 days, and practical applicability in … ting… n real samples such as tap water and urine, validating… pote… s a rapid… sitive, and reliable tool for pharmaceutic… llutant m… rin… 7]. The detection of acetaminophen usi… E modified … fur-doped reduced graphene oxide (S@r… has… onstrated e…anced electrochemical performance due to the high … rocatalytic activity of the S@rGO material. Chara… ion confirme… successful synthesis and morphology of S@… , which, when used to modify the GCE (S@rGO/GCE), significa… y improved t… current response toward acetaminophen detection co… red to the … GCE. CV and DPV studies in 0.1 M PBS (pH 7.0) rev… that S@… /GCE exhibited a low detection limit of 0.07 µM and h… y of 0.957 µA/µM·cm2, outperforming several previously reported sensors with detection limits of 1.33 µM, 0.78 µM, 0.23 … µM, 0.28 µM, 50 µM, 1.39 µM, and 3.71 µM, and sensitivities of … /µM·cm2, 0.061 µA/µM·cm2, and 0.091 µA/µM·cm2. The S@rGO/GCE sensor showed a linear current response over a wide acetaminophen concentration range (0.05 µM–91 µM in CV and 0.05 µM–125 µM in DPV), with increasing response at higher scan rates (50–500 mV/s). It also demonstrated excellent selectivity against interfering species (uric acid, glucose, cholesterol, phenol, NaCl, and dopamine at 500 µM), high repeatability over 50 cycles, and maintained stability after 15 days of storage, confirming its reliability for practical acetaminophen sensing applications [218]. Anbumannan et al. developed a sensitive and reliable electrochemical sensor for detecting acetaminophen using a GCE modified with a graphene oxide–manganese dioxide (GO–MnO_2) nanocomposite. The nanocomposite was synthesized via a simple, eco-friendly co-precipitation method and characterized for structural and functional properties. At pH 3, the sensor exhibited a strong redox response due to a two-electron, two-proton transfer mechanism between AAP and the GO–MnO_2/GCE surface. DPV revealed a wide linear detection range of 0.125–2000 µM, high sensitivity (17.04 µAμM^{-1} cm^{-2}), and a low detection limit of 7.042 nM. The sensor also demonstrated excellent anti-interference capability, reproducibility, and recovery rates between 98.47 % and 99.22 % in human urine, indicating its potential for clinical and pharmaceutical applications [219]. The detection of Naproxen using GCEs can be effectively achieved by polishing the electrode surface with commercial alumina, which enhances the electrochemically active surface area (EASA), as confirmed by AFM showing residual alumina particles. Using DPV, the alumina-polished GCE (GCE/AP) exhibits a strong electrochemical response for Naproxen, allowing for the construction of two calibration curves with linear ranges of 0.20–49.0 µmol L^{-1} and 2.92–13.5 µmol L^{-1}. The corresponding LOD and LOQ are 0.06 µmol L^{-1} and 0.20 µmol L^{-1} for the first range, and 4.0 µmol L^{-1} and 12.0 µmol L^{-1} for the second. The sensor demonstrates acceptable anti-interference performance against common tablet excipients, and when applied to pharmaceutical formulations and raw material samples, the results are

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statistically consistent with those obtained by HPLC, confirming the method's reliability and analytical accuracy [220]. Liu et al. focus on the detection of chloramphenicol (CHL) using GCEs modified with Ag/ZIF-67/g-C_3N_4. The composite electrode, Ag/ZIF-67/g-C_3N_4/GCE, exhibits excellent electrochemical properties, including a large electroactive surface area (0.343 cm2), enhanced conductivity, and low charge-transfer resistance, which are essential for sensitive and accurate detection of chloramphenicol at low concentrations. The sensor demonstrates a wide linear detection range (0.01–250.00 μmol L^{-1}) and an ultralow detection limit of 5.32 nmol L^{-1}. The incorporation of machine learning techniques, specifically backpropagation (BP) neural networks, enables real-time data analysis and improves the sensor's analytical performance. This sensor shows excellent sensitivity, selectivity (current variation within ± 8.0 % against common interferents), stability (signal decrease of only 5.55 % over 20 days), and reproducibility (RSD of 6.28 %). It performs reliably in both pharmaceutical and biological samples, achieving high recovery rates (98.00–103.00 % in pharmaceutical samples and 96.67–103.33 % in urine) with low RSD values (below 2.56 % and 2.26 %, respectively). These results demonstrate the sensor's effectiveness and robustness for chloramphenicol detection in complex sample matrices. Therefore, the developed Ag/ZIF-67/g-C_3N_4/GCE sensor provides a promising platform for sensitive and selective electrochemical detection of chloramphenicol in clinical and environmental applications [221]. The detection of Riociguat (RCG) using GCEs is exemplified by the COF-PTA@CuCoFe-LDH/GCE sensor. This advanced electrochemical sensor integrates covalent organic frameworks with a trimetallic layered double hydroxide, providing an electroactive surface area of 0.12 cm2. Structural characterization confirmed its integrity and functionality, while cyclic voltammetry and differential pulse voltammetry studies demonstrated its superior electrocatalytic activity. Optimized in PBS buffer at pH 6.0, the sensor exhibited a low LOD of 5.96 nM, a LOQ of 19.9 nM, and a wide linear range from 0.05 to 16.4 μ showed excellent repeatability and reproducibility with RSD of 2.8 and 1.85 %, respectively, and maintained selectivity with an RSD 1.15 % in the presence of common interferents. Successf lication real samples yielded recovery rates between 97.38 nd 1 % in human plasma, urine, and pharmaceutical tablet ese re ts highlight the COF-PTA@CuCoFe-LDH/GCE sensor's p tial and accurate platform for electrochemical tecti Riociguat in pharmaceutical and biological environm [222]. Q t al showed that the electrochemical detection of n using GO d sensors has excellent sensitivity and selecti par rly with a G modified GCE. This study investigates the fluence of d and oxygen content in various GO-based nanom als on their elec emical behavior. Among the materials test fluori doped GO (F GO), boron-doped partially reduced GO O), gen-doped partially reduced GO (N-rGO), and thermally re O (TrGO GO exhibited the highest electrochemical mance nap n detection. The superior performance O v ttribu higher oxygen content, which facilitated the catalytic oxidation of naproxen. Using differential pulse voltammetry (DPV), the GO/GCE sensor demonstrated a wide linear range (10 μM–1 mM), a high sensitivity of 0.60 μA/μM·cm^2, and a strong anti-interference capability. As shown in Fig. 9a, the current density increased with rising naproxen concentrations, and Fig. 9b presents a calibration plot with a strong linear relationship (R^2 = 0.9963). The sensor was also successfully tested with pharmaceutical naproxen tablets, achieving a recovery of 96.9 % with a relative standard deviation of 2.5 %, showing excellent accuracy and selectivity without interference from tablet excipients. This study highlights the critical role of oxygen functional groups and heteroatom doping in enhancing the electrochemical detection of pharmaceutical compounds like naproxen [223].

Ahmadi-Kashani et al developed a novel hemical sensor for the detection of Naproxen, using GCE mo d wit nanostructures incorporated into a graphene/polyani (PANI) na mposite. The modified electrode (PbS/graphene/PA C) exhibit significantly enhanced electrocatalytic perf ance, acteriz by reduced charge transfer resistance (Rc 6 kΩ), increa ff ve surface area (0.062 cm^2), and amplifie idation k curr owing to the synergistic effects of the co ive phene and PANI with PbS nanostructures. Electroch ical ques in g CV, EIS, and DPV confirmed the su or electro nsfe etics and electrocatalytic activity of the d electrode NAP. Under optimal pH 7.0, the sensor d nstra wide linea tection range of 1–100 μM with a low detection limit o μM, excellent repeatability (RSD ~ 2.6 %), long- bility (97 % nse retained after two weeks), and high sel vity against common iological interferents. Furthermore, the se r successfull uantified NAP in human plasma and urine samples w ecovery rat etween 98.8 % and 99.9 %, indicating its promising app ility for ical diagnostics [224]. The electrochemical detection o ressant drug Agomelatine (AG) was successfully carried out by Ansari et al using a GCE modified with a CeO_2/g-C_3N_4 mposite, where CeO_2 nanoparticles were well-dispersed on g-N_4 ets. The modified electrode, fabricated by a simple casting method, showed a significantly enhanced electroactive surface area of 0.204 cm^2, nearly eight times higher than that of the bare GCE (0.027 cm^2), and exhibited a reduced charge transfer resistance of 6.1 kΩ compared to 16.8 kΩ for bare GCE, as determined by EIS. Using CV and SWV, the modified sensor displayed a sharp anodic peak current for AG oxidation with a noticeable negative shift in peak potential (1.06 V) compared to bare GCE (1.15 V), indicating improved electrocatalytic activity. The oxidation of AG was found to be a diffusion-controlled and irreversible process involving an equal number of protons and electrons, with optimal performance at pH 6.5 in Britton-Robinson buffer. A linear response was obtained in the concentration range of 1–20 ng/mL, with a calculated LOD of 0.96 ng/mL and a LOQ of 2.92 ng/mL. The sensor also demonstrated excellent reproducibility (RSD of 0.58 % for inter-electrode and 0.56 % for intra-electrode), long-term stability (retaining 88 % of initial current after 15 days at 5 °C), and high selectivity

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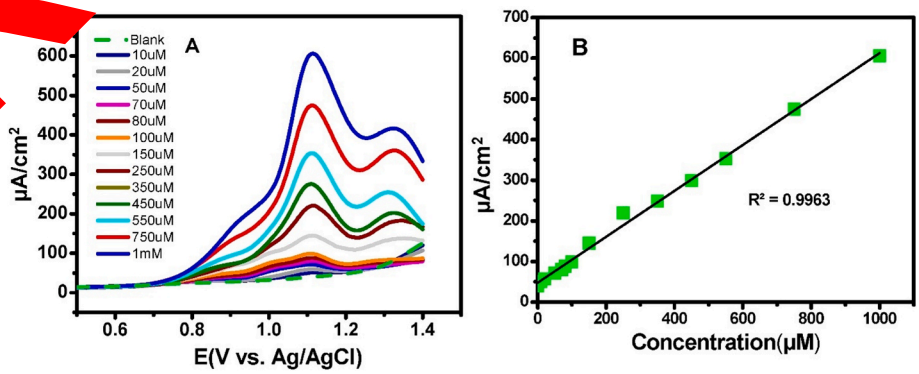


Fig. 9. (a) DPV responses at various naproxen concentrations for the GO/GCE employing DPV. (b) The calibration curve shows a linear response in current density as the concentration increases from 10 μM to 1 mM [223].

even in the presence of common excipients, ions, and other antidepressants, except for slight interference by Paroxetine. When applied to commercial tablets (Agoprex, 25 mg), the sensor achieved recovery rates between 99 and 100.1 %, confirming its reliability for pharmaceutical analysis [225]. Wang et al, developed a novel electrochemical sensor, p(L-Cys)/g-C_3N_4/GCE, by modifying a GCE with g-C_3N_4 followed by electropolymerization of 5 mM L-cysteine at pH 6.5 for 240 s. The sensor was constructed by modifying the GCE with graphitic carbon nitride (g-C_3N_4) followed by electropolymerization of 5 mM L-cysteine at pH 6.5 for 240 s. This composite structure provided a high surface area and abundant active sites, resulting in enhanced electron transfer and electrocatalytic activity. Electrochemical characterization using CV and DPV showed a notable increase in oxidation peak currents, with the p(L-Cys)/g-C_3N_4/GCE delivering peak currents approximately 1.5, 3.6, and 8 times greater than those from p(L-Cys)/GCE, g-C_3N_4/GCE, and bare GCE, respectively. The sensor demonstrated an excellent linear detection range of 7.5–780 μM, and a LOD of 6 nM, along with a high sensitivity of 0.11841 μA/μM. In terms of stability, the sensor retained 96.3 % of its response after 30 days, and exhibited a minimal signal decrease of 4 % after 90 consecutive scans—superior to other configurations. It also showed excellent reusability with an RSD of 3.10 % across four identical electrodes. Selectivity tests confirmed strong resistance to interference from other substances, and validation with pharmaceutical samples and serum from five breast cancer patients aged 35–50 years yielded recovery rates above 97.20 % and RSDs below 5.11 %, with results comparable to ELISA. Overall, the sensor exhibits a reliable and robust platform for precise MTX monitoring in clinical and pharmaceutical settings, with competitive LOD and linear range performance compared to previously reported electrochemical MTX sensors [226]. Prasad et al developed a highly sensitive electrochemical sensor for pharmaceutical detection based on a GCE modified through the e... tropolymerization of L-threonine (Thr), followed by the deposit... GO and MWCNT. The resulting MWCNT/GO/Poly (Thr)/GCE compo... exhibited superior surface properties, as confirmed by FE-SEM, Ram... and ATR-IR analyses. Electrochemical characterizat... using [F... $(CN)_6]^{3-}/^{4-}$ as a redox probe showed a progressive ...ea... charge transfer resistance (Rct) from 432 Ω (bare GCE) to ... Ω (MW...NT/GO/Poly (Thr)/GCE), and an increase in electroact... urfac... 0.221 cm^2. The sensor demonstrated enhan... ele...lytic perfo... mance, delivering peak current response... out 15 and ...imes higher than the bare GCE for cathodic and ... processes o... respectively. DPV revealed a linear detec... ran...om 5 to 200 μM with a low LOD of 0.16 μM, and the ...ression equ... Ipa (μA) = 1.56 + 0.131[PCT] (μM) (R^2 = 0.99... The electrode s... d a strong correlation with the square roo... scan rat... indicating a diffusion-controlled redox process, with slo... lues ...68 and 0.59 from log–log plots of scan rate vs. current. It als... strated h... selectivity toward PCT in the presence of n... ephri... d folic..., with minimal interference from other io... nd ...olecu... at 100-fold excess. The sensor maintained 96.5 % of its initial response after 7 days, with repeatability and reproducibility reflected by RSD values of 3.5 % and 2.8 %, respectively. Moreover, the sensor was successfully applied in analyzing PCT in tablet and human serum samples using the standard addition method [227]. Muungani et al prepared an electrochemical sensor based on a calcium copper tetrasilicate ($CaCuSi_4O_{10}$) modified GCE was fabricated for the sensitive and selective detection of the antibacterial drug norfloxacin using CV and DPV. The $CaCuSi_4O_{10}$/GCE showed enhanced electrochemical properties, with a lower charge transfer resistance (22.1 Ω·cm^2) compared to the bare GCE (43.5 Ω·cm^2), and a larger electroactive surface area (0.0463 cm^2 vs. 0.0384 cm^2), resulting in improved electron transfer kinetics. Electrochemical behavior studies using $[Fe(CN)_6]^{3-}/^{4-}$ as a redox probe co... this enhancement. Optimization of detection conditions reve... that p... in PBS yielded the highest peak current (3.406 μA) f... rfloxacin o... tion, as illustrated in Fig. 10(a). The corresponding p... potential (E... was 1.067 V, and varied linearly with p... follow... the e... tion Ep = 1.33–0.05543pH with a corr... on coefficie... 9484 (Fig. 10 (b)), indicating a proton-c... ed ele... n transf... ocess. The sensor displayed two linear det... n ra... s (0.01–0.55 μM and 0.55–82.1 μM), and a low LO... f 0.0... M. It m... ained selectivity in the presence of interf... nts like u... Na^+, ..., $Mg2^+$, and SO_4^{2-}, with a retained signa... 8 % and a... of 2.2 %. The sensor also demonstrate... xcell... reproducibili... (RSD = 3.14 %) and long-term stability (2.37 % devi... after two weeks). Application to a real pharm... al sample (... 400) yielded recovery values between 98... and 102.1 %, confirming the sensor's practical reliability for dr... analysis [228].

...e electroch... cal detection of gemcitabine (GMB) using GCE offers ...mple, fas... nd reliable method for pharmaceutical formulation contro... optimized DPV and amperometric techniques with a boron-doped diamond electrode (BDDE), which provides a broad po... window, allowing for the detection of GMB. The DPV method ...owe... linear range from 2.5 μg/mL to 50 μg/mL, with a LOD of 0.85 μg/mL, while the amperometric method exhibited a wider linear range from 0.5 μg/mL to 65 μg/mL, with a LOD of 0.15 μg/mL. The electrochemical oxidation of gemcitabine was found to be irreversible and diffusion-controlled. Both methods demonstrated excellent performance for pharmaceutical formulations, with reasonable recovery rates and no significant interference from excipients. Real sample analysis was conducted using GMB hydrochloride powder for infusion solutions, with concentrations of 2.5, 10, and 25 μg/mL prepared in PBS (pH 7.4; 0.05 M) and tested using the optimized DPV procedure (Fig. 11a). Similarly, amperometric analysis was performed by adding measured volumes of pharmaceutical stock solution to PBS (pH 7.4; 0.05 M) under continuous stirring, as shown in Fig. 11b. These electrochemical methods provide a low-cost, rapid, and stable alternative to traditional techniques such as HPLC for the analysis of gemcitabine in pharmaceutical products [229].

The detection of ciprofloxacin (CP) and acetaminophen has been

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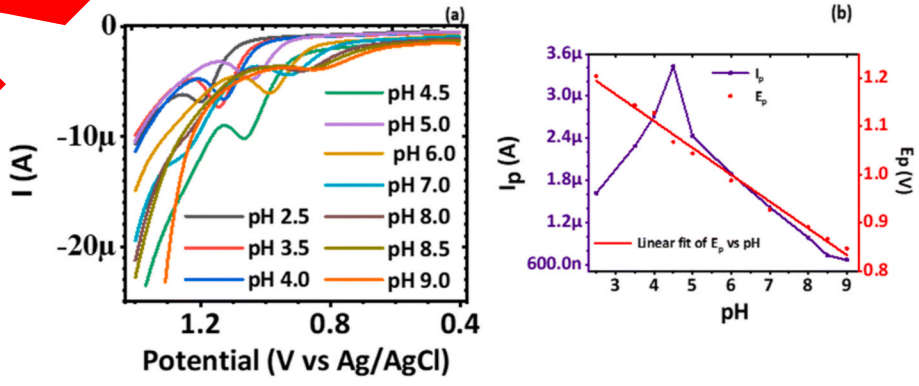


Fig. 10. Plots of (a) I vs. V, (b) ip vs. pH, and Ep vs. pH from differential pulse voltammetry of 0.1 mM norfloxacin solution at the $CaCuSi_4O_{10}$/GCE [228].

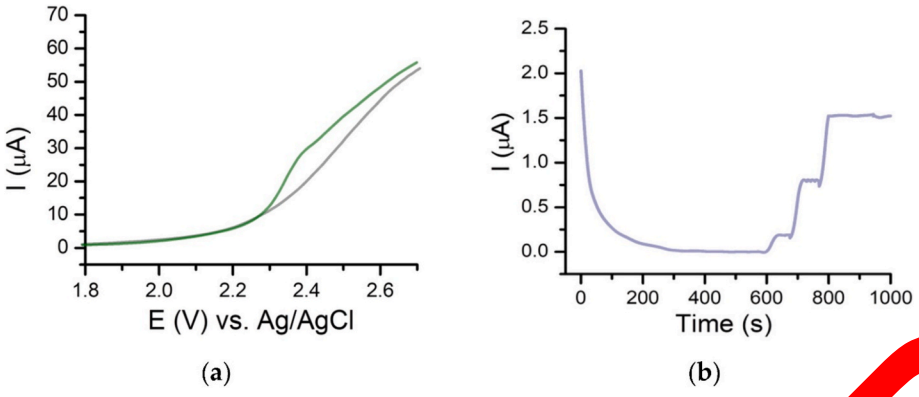


Fig. 11. (a) DPVs in a pH 7.4 PBS solution (gray) and a 10 μg/mL GMB infusion solution (green); (b) Chronoamperogram of 25, and 140 μL additions of 500 μg/mL GMB infusion solution in 5 mL PBS (pH 7.4; 0.05 M), which correspond to 2.5, 10, and 25 μg/mL GMB [229].

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advanced through the development of an ecofriendly and cost-effective electrochemical sensor based on reduced graphene oxide doped with cerium magnesium zinc sulfide (rGO@CMZ) nanocomposites. This sensor demonstrates high sensitivity with low detection limits of 71 nM for CP and 27 nM for AP within a linear concentration range of 1 μM to 50 μM at neutral pH (pH 7). Comprehensive characterization of the rGO@CMZ nanocomposite by techniques including XRD, FE-SEM, TEM, XPS, EDX, thermogravimetric analysis, and BET and BJH analysis confirmed its structural and conductive properties. The modified glassy carbon electrode (GCE/rGO@CMZ) showed a significantly increased electroactive surface area of 1.49 cm^2, compared to 0.88 cm^2 for GCE/CMZ and 0.36 cm^2 for the bare electrode, along with superior electrical conductivity. Additionally, the sensor exhibited excellent reproducibility, selectivity, and long-term stability, enabling reliable detect CP and AP in complex biological samples such as urine and sa indicating its strong potential for practical environmental and clin monitoring of pharmaceutical pollutants [230]. Yari et al resented innovative electrochemical sensor based on a poro posite Fe_3O_4#GO-PCA, where Fe_3O_4 nanoparticles are edded a poly-citric acid-coated graphene oxide matrix, servi n effi sing platform for detecting Piroxicam (PX), a c mo inflammatory drug. Characterization tec ues suc UV–vis spectrophotometry, FE-SEM, FT-IR, and X firmed the ssful synthesis and structural features of th an posite. Elec chemical studies using CV and EIS revealed a signifi nhancement in conductivity, reducing resistance 1942 Ω on ba E to less than 80 Ω on the Fe_3O_4#GO-PCA mo d elec rode, which facilitated improved electron transfer and re ble red behavior. The modified electrode showed a linear detectio PX fro 0.04 to 70.0 μg/mL with a low detection limit of 0.02 at pH and exhibited a diffusion-controlled ele tion It demonstrated excellent selectivity a st con n interfe species, rapid response time (~1 min), and repro ility and repeatability with relative standard deviations o and , respectively. Importantly, the sensor was successfull lied for PX quantification in pharmaceutical tablets, human blood pla and urine, achieving recovery rates between 92.20 % and 99.0 , comparable or superior to standard HPLC methods. Overall, the Fe_3O_4#GO-PCA modified electrode proves to be a highly sensitive, accurate, and reliable tool for voltammetric detection of PX in pharmaceutical and clinical samples [231]. Garkani Nejad et al developed a novel electrochemical sensor for tramadol detection using a GCE modified with a nanocomposite of UiO-66-NH2 MOF and third-generation poly(amidoamine) (G3-PAMAM) dendrimer. The sensor exhibited excellent electrocatalytic properties for tramadol oxidation, confirmed through various characterization techniques such as XRD, FE-SEM, and FT-IR spectroscopy. The sensor demonstrated a wide linear dynamic range for tramadol detection from 0.5 μM to 500 μM, with a low LOD of 0.2 μM. The electrochemical response showed a substantial increase in oxidation cur with h her tra concentrations. Specifically, as illustrate Fig. the DPV response revealed an apparent increase in a x curren pa) with increasing tramadol concentrati and the ration rve showed a linear relationship with th ation Ipa = × tramadol + 0.7839 (R^2 = 0.9997). The so ctively disti ished tramadol from acetaminophen, with separate tion potentials of 780 mV for tramadol and 370 m cetaminop The electrochemical behavior followed a diff n-controlled process, nd the sensor displayed high stability, re tability (4.1 % RSD), and reproducibility (2.8 % RSD). Additionally, the sensor s essfully analyzed tramadol and acetaminophen in ph ceutical f ulations, with recovery rates between 96.7 % and 103.3 ighl ng its practical applicability in drug analysis [232].

Below comparative table (Table 3) summarizing the key features e electrochemical sensors described for the detection of various utical pollutants. The table includes the target analyte, electrode modification, detection technique, linear range, LOD, sensitivity, stability, selectivity, and real-sample application performance.

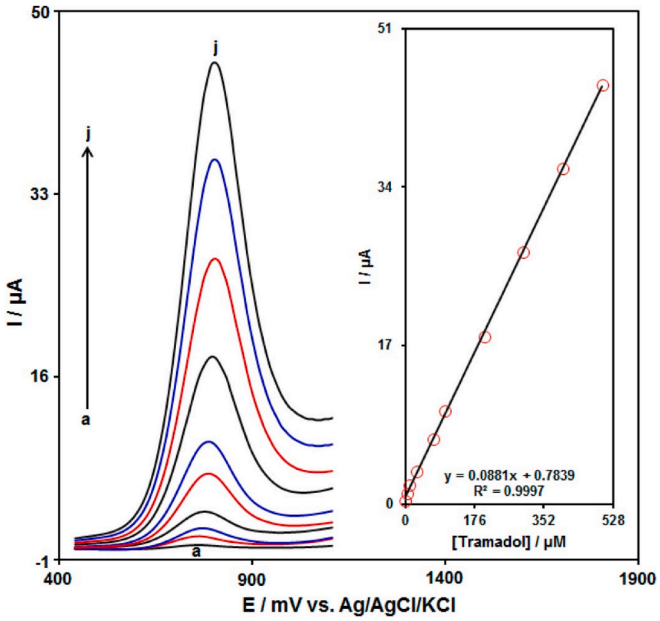


Fig. 12. DPVs obtained for the UiO-66-NH_2 MOF/PAMAM-modified GCE exposed to different tramadol concentrations (a: 0.5 μM, b: 5.0 μM, c: 10.0 μM, d: 30.0 μM, e: 70.0 μM, f: 100.0 μM, g: 200.0 μM, h: 300.0 μM, i: 400.0 μM, and j: 500.0 μM)) in PBS (pH 7.0, 0.1 M). Insets: Correlations between the Ipa and tramadol levels [232].

Table 3
Comparison of key features of the electrochemical sensors for the detection of various pharmaceutical pollutants.

Sensor Material	Target Pharmaceutical	Linear Range (μM)	LOD (μM)	Sensitivity (μA/μM·cm^2)	Electroactive Surface Area (cm^2)	Charge Transfer Resistance (Ω)	Selec[illegible]	[illegible]ability	Reproducibility (RSD)	Reference
rGO/GCE	Acetaminophen	5–800	0.0021	Not specified	Not specified	Not specified	[illegible] specifi[illegible]	Not specified	Not specified	[216]
NiCoFe LDH@g-CN/GCE	Acetaminophen	0.166–1225.5	0.06	2.4338	Not specified	418	High against [illegible]n interferents	Stable for 15 days	2.3–2.5 %	[217]
S@rGO/GCE	Acetaminophen	0.05–125	0.07	0.957	Not specified	Not specified	against uric ac[illegible], glucose, c[illegible]rol, phenol, NaCl, dopa[illegible](500 μM)	Stable for 15 days	Not specified	[218]
GO–MnO_2/GCE	Acetaminophen	0.125–2000	0.007042	17.04	Not specified	[illegible]ied	Excellen[illegible]-interference	Not specified	Not specified	[219]
Alumina-polished GCE (GCE/AP)	Naproxen	0.20–49.0, 2.92–13.5	0.06	Not specified	Enhanced by AFM	[illegible]ot specified	Acceptable against tablet excipients	Not specified	Not specified	[220]
Ag/ZIF-67/g-C_3N_4/GCE	Chloramphenicol	0.01–250	0.00532	Not specified	0.343	[illegible]ow (not [illegible]ntified)	Current variation within ± 8.0 %	5.55 % signal decrease (20 days)	6.28 %	[221]
COF-PTA@CuCoFe-LDH/GCE	Riociguat	0.05–16.4	0.00596	Not specified	0.12	No[illegible]	High, RSD 1.15 % with interferents	Not specified	1.85–2.81 %	[222]
GO/GCE	Naproxen	10–1000	Not specified	0.60	Not s[illegible]ified	[illegible]t specified	High, no interference from excipients	Not specified	2.5 %	[223]
PbS/graphene/PANI/GCE	Naproxen	1–100	0.08	Not specified	0.062	6000	High against biological interferents	97 % response after 2 weeks	~2.6 %	[224]
CeO_2/g-C_3N_4/GCE	Agomelatine	0.001–0.02 (ng/mL)	0.00096 (ng/mL)	Not s[illegible]ed	[illegible]204	6100	High, slight interference from Paroxetine	88 % response after 15 days	0.56–0.58 %	[225]
p(L-Cys)/g-C_3N_4/GCE	Methotrexate	7.5–780	0.006	[illegible]11	[illegible]cified	Not specified	Strong resistance to interferents	96.3 % response after 30 days	3.10 %	[226]
MWCNT/GO/Poly (Thr)/GCE	Acetaminophen	5–200	0.16	Not sp[illegible]	0.221	115	High against norepinephrine, folic acid	96.5 % response after 7 days	2.8–3.5 %	[227]
$CaCuSi_4O_{10}$/GCE	Norfloxacin	0.01–0.55, 0.55–82.1	0.00[illegible]	Not specifie[illegible]	0.0463	22.1 (Ω·cm^2)	High, 97.8 % signal retention	2.37 % deviation after 2 weeks	3.14 %	[228]
BDDE	Gemcitabine	0.5–65 (amperometri[illegible]	0.15	[illegible]ecified	Not specified	Not specified	No significant interference from excipients	Not specified	Not specified	[229]
rGO@CMZ/GCE	Ciprofloxacin, Acetaminophen	1–50	0.0[illegible](CP), [illegible] (AP)	Not specified	1.49	Not specified	Excellent selectivity	Excellent long-term stability	Not specified	[230]
Fe3O4#GO-PCA/GCE	Piroxicam	0.04–70 (μ[illegible]	[illegible].02 (μg/m[illegible]	Not specified	Not specified	<80	Excellent against common interferents	Not specified	1.2–3.1 %	[231]
UiO-66-NH2/G3-PAMAM/GCE	Tramadol	[illegible]2	[illegible]	Not specified	Not specified	Not specified	High, distinguishes from acetaminophen	High stability	2.8–4.1 %	[232]

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8. Challenges and limitations

Despite significant advancements in the development of modified GCEs for the electrochemical sensing of pharmaceutical pollutants, several challenges and limitations persist that hinder their practical application and widespread adoption. One of the foremost challenges lies in the reproducibility of electrode modification processes. The fabrication of nanostructure-modified GCEs often involves complex, multi-step synthesis procedures that are sensitive to variations in experimental conditions such as temperature, pH, and precursor concentration. This results in inconsistent electrode performance, making it difficult to achieve reproducible and scalable sensor fabrication. Additionally, the stability of the nanomaterials used for electrode modification is a critical issue. Nanostructures, particularly metal nanoparticles and metal oxides, are prone to aggregation, leaching, or degradation under electrochemical and environmental stress, which can compromise sensor longevity and reliability.

Another significant limitation is related to the analytical performance of the sensors. Although nanostructure modification generally enhances sensitivity, issues with selectivity remain prevalent. In real sample matrices, the presence of interfering species such as other pharmaceuticals, ions, and organic compounds can lead to cross-reactivity and false signals. Moreover, some sensors still exhibit detection limits that are insufficient for detecting pharmaceutical pollutants at the trace levels required by environmental monitoring guidelines. Signal drift over time, caused by fouling or changes in the electrode surface, further complicates accurate quantification, especially in long-term applications.

The translation of these sensors from laboratory research to real-world applications is also hindered by challenges associated with analyzing real samples. Environmental samples such as wastewater or surface water often contain complex matrices that can suppress or distort electrochemical signals, necessitating extensive pre-treatment steps such as filtration or pH adjustment. These additional steps reduce the practicality of the sensors for in situ or on-site monitoring applications. Furthermore, most studies remain at the proof-of-concept stage, and few have demonstrated effective integration of these sensors into portable or automated systems suitable for field deployment.

Material-specific limitations also pose significant hurdles. While advanced nanomaterials, such as graphene derivatives, carbon nanotubes, and metal nanoparticles, offer enhanced electrochemical properties, their high cost, limited availability, and environmentally harmful synthesis routes restrict their large-scale implementation. Additionally, concerns regarding the toxicity and environmental impact of nanomaterials, particularly in the case of metal-based nanoparticles, raise questions about the long-term sustainability of their use in environmental sensors.

From a broader perspective, the lack of standardization across studies further complicates progress in the field. Variations in experimental parameters, such as supporting electrolytes, scan rates, and operating pH—make it difficult to compare the performance of different sensors or replicate results across laboratories. Moreover, long-term stability and durability studies are often lacking, with most reports focusing on short-term performance metrics. This gap in long-term evaluation leaves the practical lifespan and operational reliability of the sensors largely unverified.

Finally, there are notable barriers to the integration and commercialization of these sensing platforms. Although the concept of coupling electrochemical sensors with Internet-of-Things (IoT) devices and wireless monitoring systems is gaining interest, practical implementations remain limited. The transition from lab-scale fabrication to mass production also faces technical and economic challenges. Additionally, meeting the stringent regulatory standards for environmental monitoring and obtaining necessary certifications for commercial deployment remain unresolved issues.

9. Future perspectives and research directions

Despite substantial progress in the field of electrochemical sensing using nanostructure-modified GCEs, numerous challenges and research opportunities remain that can drive the next wave of innovation. To enhance the practical applicability, selectivity, and sensitivity of these sensors for pharmaceutical pollutants, future research should focus on the following areas:

One of the most promising directions is the design and development of advanced hybrid nanomaterials. While individual nanostructures such as graphene, carbon nanotubes, and metal nanoparticles have shown excellent electrochemical properties, their combination into hybrid systems can offer synergistic effects. For instance, coupling carbon-based nanomaterials with metal oxides or conducting polymers can improve electron transfer kinetics, increase surface area, and provide specific active sites for analyte interaction. Doping strategies, such as nitrogen-doping of graphene or the introduction of heteroatoms into carbon frameworks, have also been shown to enhance electrochemical activity. Future efforts should focus on tailoring these hybrid nanostructures at the molecular level using techniques like atomic layer deposition, templating, or electrochemical synthesis to engineer specific functionalities relevant to pharmaceutical detection.

Equally important is the development of green, sustainable, and scalable synthesis methods for these nanomaterials. Current fabrication techniques often involve toxic solvents, expensive precursors, and high-temperature conditions that are not environmentally friendly or scalable. Future work should explore green chemistry approaches, such as biosynthesis using plant extracts, hydrothermal or solvothermal methods with water-based solvents, and solvent-free mechanochemical synthesis. Microwave- and ultrasound-assisted synthesis methods also show promise in reducing reaction times and energy consumption. These approaches not only reduce the environmental footprint but also make it feasible to produce nanomaterials at the industrial scale, which is essential for the large-scale deployment of electrochemical sensors.

Selectivity remains a critical bottleneck in the development of electrochemical sensors for pharmaceutical pollutants, particularly when applied to complex environmental matrices such as wastewater and surface water. Future sensor designs should incorporate selective recognition elements, such as molecularly MIPs, aptamers, or enzyme-mimetic materials to improve target specificity. MIPs, for example, can be synthesized to possess binding cavities that are complementary in shape, size, and functional groups to a specific pharmaceutical compound. Aptamer-functionalized electrodes can offer high affinity and selectivity towards particular drug molecules. Additionally, anti-fouling strategies should be integrated into sensor architectures to prevent biofouling and surface passivation. This can include the use of zwitterionic polymers, PEGylation, or the incorporation of superhydrophilic and self-cleaning surfaces, all of which can enhance the durability and reliability of sensors in real-world environments.

Another key area of development is the miniaturization of electrochemical sensing platforms and their integration with portable and wearable technologies. With growing interest in decentralized and in-situ monitoring, there is a need to design flexible, lightweight, and low-power sensors that can be embedded in handheld or wearable devices. Glassy carbon electrodes can be miniaturized using screen-printing or inkjet-printing techniques, and when modified with nanostructures, they can maintain high performance at a microscale. The incorporation of microfluidic systems allows for on-chip sample processing and controlled reagent delivery. Furthermore, integrating sensors with wireless communication technologies, such as Bluetooth or near-field communication (NFC), can enable real-time data acquisition and transmission to smartphones or cloud-based platforms, facilitating environmental surveillance and early warning systems for pharmaceutical contamination.

Real-world validation of electrochemical sensors is another critical research priority. Many studies still focus on idealized laboratory

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conditions using buffer solutions and standard pharmaceuticals, which do not reflect the complexity of environmental samples. Future research must emphasize the application of sensors in real samples, such as municipal wastewater, hospital effluents, and natural waters, to assess their performance in the presence of co-contaminants and varying ionic strengths. The development of standardized protocols for sensor calibration, analyte recovery, and data interpretation is essential to improve reproducibility and comparability across studies. Furthermore, interlaboratory collaborations and round-robin testing should be encouraged to establish consensus on performance benchmarks and align research outcomes with regulatory frameworks, such as those from the U.S. EPA, the EU Water Framework Directive, or WHO guidelines.

The integration of artificial intelligence (AI) and machine learning (ML) into electrochemical sensing platforms represents an exciting and underexplored frontier. Electrochemical signals, particularly those from voltammetry or impedance spectroscopy, often contain rich and complex information. AI and ML can be applied to automate the interpretation of such signals, identify hidden patterns, and enhance the detection of low-abundance pharmaceuticals. For example, supervised learning algorithms can be trained on known datasets to classify and quantify multiple pharmaceuticals simultaneously, even in the presence of overlapping signals. Unsupervised learning and clustering techniques can also be used for anomaly detection or to monitor sensor drift. Furthermore, adaptive algorithms can enable self-calibrating and self-correcting sensors that improve in performance over time, making them more robust for long-term environmental deployment.

Sensor stability and reusability are significant factors determining the practical usability and cost-effectiveness of electrochemical sensors. Modified GCEs often suffer from degradation of active sites, leaching of nanostructures, and surface fouling, especially during prolonged use in complex samples. Future research should focus on improving the adhesion and robustness of nanostructured modifiers using techniques such as covalent bonding, polymer encapsulation, or electropolymerization. Additionally, developing regenerative sensing surfaces that can be electrochemically cleaned or thermally reactivated would allow sensors to be reused multiple times without significant loss in performance. Materials that exhibit intrinsic self-cleaning or anti-fouling properties could also play a critical role in extending sensor lifespans, especially in harsh or highly polluted environments.

Finally, the dynamic nature of pharmaceutical pollution, characterized by the continuous emergence of new compounds and transformation products, necessitates the development of broad-spectrum and multiplexed sensing systems. Future research should not only focus on detecting known pharmaceuticals but also on identifying emerging contaminants such as metabolites, by-products of degradation, and novel synthetic drugs. Multi-analyte detection platforms, such as electrode arrays modified with different nanostructures or universal sensing layers, can offer comprehensive monitoring capabilities. Moreover, understanding the synergistic or antagonistic interactions among co-existing pollutants is crucial, as such interactions can significantly influence electrochemical responses and environmental risk assessments. Developing sensors capable of capturing these complex dynamics will be essential for a more accurate and holistic understanding of pharmaceutical pollution in the environment.

10. Conclusion

The presence of pharmaceutical pollutants in aquatic environments poses a growing threat to ecosystems and public health, necessitating the development of sensitive, selective, and reliable detection strategies. Electrochemical sensing—particularly using GCEs modified with nanostructures—has emerged as a highly promising approach due to advantages such as low cost, rapid response, ease of miniaturization, and the feasibility of on-site monitoring. This review has introduced recent advances in GCE modification strategies, offering a novel perspective by systematically analyzing how various nanostructured materials—including carbon-based nanomaterials, metal nanoparticles, metal oxides, conductive polymers, and hybrid composites—enhance electrochemical sensor performance for pharmaceutical pollutant detection. We emphasized the mechanisms of signal amplification and the critical influence of nanomaterial properties on improving sensitivity, selectivity, and stability. A key novelty of this review is its focus on recent engineering approaches to GCE modification, providing a comprehensive understanding of how tailored nanostructures can be leveraged to design next-generation sensing platforms. Moreover, these nanostructures also hold significant potential in photocatalysis, offering benefits such as enhanced light absorption, efficient charge separation, and increased surface area for catalytic reactions. These features can further support dual-function systems that simultaneously detect and degrade pharmaceutical pollutants. Despite notable advancements, several challenges remain, including limited long-term sensor stability, selectivity in complex matrices, scalability of nanomaterial production, and the absence of standardized testing protocols. Nevertheless, the integration of cutting-edge nanomaterials with green synthesis techniques, artificial intelligence, and flexible electronics paves the way for robust and intelligent electrochemical sensing systems. In conclusion, nanostructure-modified GCEs stand out as versatile and powerful platform for environmental monitoring. Ongoing interdisciplinary research across materials science, analytical chemistry, environmental engineering, and data science will be essential to overcome existing barriers and fully harness the potential of these advanced sensors for protecting environmental and public health.

CRediT authorship contribution statement

Emmanuel Kipkorir: Writing – review & editing, Writing – original draft, Conceptualization. **Otieno Kimani:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] B. Bai, T. Xu, Q. Nie, P. Li, Temperature-driven migration of heavy metal Pb2+ along with moisture movement in unsaturated soils, Int. J. Heat Mass Transf. 153 (2020) 119573.
[2] X. Li, X.-L. Shi, J. Wang, K. Shi, Q. Wang, Effect of different hydrogen donors on the catalytic conversion of levulinic acid to γ-valerolactone over non-noble metal catalysts, J. Ind. Eng. Chem. (2024).
[3] B. Bai, F. Bai, X. Li, Q. Nie, X. Jia, H. Wu, The remediation efficiency of heavy metal pollutants in water by industrial red mud particle waste, Environ. Technol. Innov. 28 (2022) 102944.
[4] Z.Q. Yang, K.P. Hou, T.T. Guo, Study on the effects of different water-cement ratios on the flow pattern properties of cement grouts, Appl. Mech. Mater. 71 (2011) 1264–1267.
[5] Z.Q. Yang, K.P. Hou, T.T. Guo, Research on time-varying behavior of cement grouts of different water-cement ratios, Appl. Mech. Mater. 71 (2011) 4398–4401.
[6] Y.-Z. Wu, J. Wang, Y.-H. Hu, Q.-S. Sun, R. Geng, L.-N. Ding, Antimicrobial Peptides: Classification, Mechanism, and Application in Plant Disease Resistance, Probiotics Antimicrob. Proteins (2025) 1–15.
[7] X. Sui, Q. Chen, G. Gu, Adaptive grayscale adjustment-based stripe noise removal method of single image, Infrared Phys. Technol. 60 (2013) 121–128.
[8] J. Sheng, R. Zhang, H. Yang, Inter-basin water transfers and water rebound effects: The South-North water transfer Project in China, J. Hydrol. 638 (2024) 131516.
[9] B. Bai, J. Chen, F. Bai, Q. Nie, X. Jia, Corrosion effect of acid/alkali on cementitious red mud-fly ash materials containing heavy metal residues, Environ. Technol. Innov. 33 (2024) 103485.

RETRACTED

[10] G.-H. Tian, K. Sun, P. Huang, C.-M. Zhou, H.-J. Yao, Z.-J. Huo, H.-F. Hao, L. Yang, C.-S. Pan, K. He, Long-term stimulation with electroacupuncture at DU20 and ST36 rescues hippocampal neuron through attenuating cerebral blood flow in spontaneously hypertensive rats, Evidence-Based Complementary and Alternative Medicine 2013 (2013) 482947.

[11] X. Sui, Q. Chen, G. Gu, Adaptive bias voltage driving technique of uncooled infrared focal plane array, Optik 124 (2013) 4274–4277.

[12] X. Sui, Q. Chen, L. Bai, Detection algorithm of targets for infrared search system based on area infrared focal plane array under complicated background, Optik 123 (2012) 235–239.

[13] Q. Wang, Z. Lu, D. Wang, K. Wang, Mechanosensor for proprioception inspired by ultrasensitive trigger hairs of Venus flytrap, Cyborg Bionic Syst. 5 (2024) 0065.

[14] J. Cai, W. Wang, D. Guo, Q. Yang, Q. Yan, Adaptive backstepping control for a class of nonlinear systems with output modeling error and external disturbance, Trans. Inst. Meas. Control (2025), 01423312251318793.

[15] T. Wang, H.R. Karimi, H. Wang, N. Xu, L. Li, X. Zhao, A reinforcement learning methodology to hierarchical sliding-mode surface H∞ H_∞ control of nonlinear systems via a dynamic event-triggered mechanism, Asian J. Control (2025).

[16] Y. Liu, Y. Wang, Y. Fu, N.N. Wang, S.Q. Zhan, L. Sheng, H.Y. Yang, C. Liu, Healable and transparent ionic conductive hydrogels based on PNATF as multiple-signal sensors, ACS Appl. Polym. Mater. 7 (2025) 2529–2540.

[17] Z. Gao, N. Xu, G. Zong, H. Wang, B. Niu, X. Zhao, Fuzzy weight-based secure formation control for two-order heterogeneous multi-agent systems via reinforcement learning, Inf. Sci. 698 (2025) 121782.

[18] Y. Zhiquan, Q. Shanguang, H. Kepeng, Time-dependent Behavior Characteristics of Power-law Cement Grouts Applied in Geotechnical Engineering, Electron. J. Geotech. Eng. 21 (2016) 1017–1024.

[19] Y. Wang, X. Zhang, Y. Ju, Q. Liu, Q. Zou, Y. Zhang, Y. Ding, Y. Zhang, Identification of human microRNA-disease association via low-rank approximation-based link propagation and multiple kernel learning, Front. Comp. Sci. 18 (2024) 182903.

[20] J. Sheng, Q. Cheng, H. Yang, Water markets and water inequality: China's water rights trading pilot, Socioecon. Plann. Sci. 94 (2024) 101929.

[21] X. Sui, L. Bai, Q. Chen, G. Gu, Influencing factors of microscanning performance based on flat optical component, Chin. Opt. Lett. 9 (2011) 052302.

[22] Y. Wu, H. Liang, N. Zhao, B. Niu, Low-computation-based adaptive self-triggered bipartite consensus control for nonlinear multi-agent systems subject to sensor faults, IEEE Trans. Control Network Syst. (2024).

[23] W. Zhengzhi, L. Ming, J. Xiuqin, Changes of neuronal nitric oxide synthase in relevant cerebral regions in spontaneous senile dementia model and regulation of Tiantai I, Chin. J. Tissue Eng. Res. (2005) 244–247.

[24] T. Zhang, J. Ou-Yang, X. Yang, B. Zhu, Transferred PMN-PT Thick Film on Conductive Silver Epoxy, Materials 11 (2018) 1621.

[25] S.X. Yang, B. Liu, M. Tang, J. Yang, Y. Kuang, M.Z. Zhang, C.Y. Zhang, C.Y. Wa… J.C. Qin, L.P. Guo, Extraction of flavonoids from Cyclocarya paliurus (Juglandaceae) leaves using ethanol/salt aqueous two-phase …pled wit… ultrasonic, J. Food Process. Preserv. 44 (2020) e14469.

[26] L. Zhao, W. Weng, M. Ni, H. Shen, S. Zhang, Y. Chen, …, L. Fan, …Mao, L. Qin, Rubidium salt can effectively relieve the sy… of DSS…ced ulcerative colitis, Biomed. Pharmacother. 181 (2024) …4.

[27] L. Zhao, X. Wang, J. Li, X. Tan, L. Fan, Z. Zha… Leng… Cyclocarya paliurus on hypoglycemic effect in type 2 d… mice, M… Monitor: Int. Med. J. Exp. Clin. Res. 25 (2019) 2976.

[28] B.P. Zhu, W.K. Guo, G.Z. Shen, Q. Zh… g, Structure a… trical properties of (111)-oriented Pb (Mg… Nb 2… -PbZrO 3-PbTi… 3 thin film for ultra-high-frequency transdu… applications, …Trans. Ultrason. Ferroelectr. Freq. Control 58 (…) 1962–1967.

[29] L.-C. Zhao, Y. He, X. Deng, … Xia, J. Liang, G.-L. Y… Y. Li, H. Wang, Ultrasound-assisted extr… of syri… from the bark of Ilex rotunda thumb using response surface… odolog… J. Mol. Sci. 13 (2012) 7607–7616.

[30] 隋修宝, 柏连发, 陈钱, 顾… …hm for eli…nating stripe noise in infrared image, Journal of Infrared … limeter W… 31 (2012) 106–112.

[31] X. Sui, Q. Ch… N. Liu, … se m… of resistance-type microbolometer, Opt. Rev. … 5–531.

[32] B. Zhu… i, C. W… Y. Zhu, X… H. Zheng, Q. Zhou, K.K. Shung, Self-focu… ScN film… asound transducer for individual cell manipulation, ACS Senso… 017…

[33] B.P. Zhu… u, K.K. Shung, Q. Wei, Y.H. Huang, Sol-gel derived PMN-PT thick film h… frequency ultrasound transducer applications, in: 2009 IEEE International … nics Symposium, IEEE, 2009, pp. 2197–2200.

[34] X. Sui, J. Zeng, Q… G. Gu, High spatial resolution recording of near-infrared hologram based on …oto-induced phase transition of vanadium dioxide film, Opt. Lett. 40 (2015) 1595–1598.

[35] X. Sui, W. He, C. Zuo, Q. Chen, G. Gu, 10.6 μm Infrared light photoinduced insulator-to-metal transition in vanadium dioxide, Infrared Phys. Technol. 64 (2014) 103–107.

[36] X. Xiang, H. Liu, C. Zheng, N. Jiang, F. Huang, Q. Zhou, Flavor Profile of 4-Iso-thiocyanato-1-butene in Microwave Rapeseed Oil and Its Anti-Inflammatory Properties In Vitro, J. Agric. Food Chem. (2025).

[37] X. Yi, R. Zhao, Y. Lin, The impact of nighttime car body lighting on pedestrians' distraction: A virtual reality simulation based on bottom-up attention mechanism, Saf. Sci. 180 (2024) 106633.

[38] B. Zhu, J. Han, J. Shi, K.K. Shung, Q. Wei, Y. Huang, M. Kosec, Q. Zhou, Lift-off PMN–PT thick film for high-frequency ultrasonic biomicroscopy, J. Am. Ceram. Soc. 93 (2010) 2929–2931.

[39] B. Zhu, D. Wu, Y. Zhang, J. Ou-Yang, S. Chen, X. Yang, Sol–gel derived PMN–PT thick films for high frequency ultrasound linear array applications, Ceram. Int. 39 (2013) 8709–8714.

[40] Y. Han, Q. Xu, J.-N. Hu, X.-Y. Han, W. Li, L.-C. Zhao, Maltol, a food flavoring agent, attenuates acute alcohol-induced oxidative damage in mice, Nutrients 7 (2015) 682–696.

[41] X. Sui, Q. Chen, G. Gu, A novel non-uniformity evaluation metric of infrared imaging system, Infrared Phys. Technol. 60 (2013) 155–160.

[42] Y. Zhiquan, Z. Yingyan, Q. Shanguang, H. Kepeng, Types and space distribution characteristics of debris flow disasters along China-Pakistan Highway, Electron. J. Geotech. Eng. 21 (2016) 191–200.

[43] Z. Yu, Z. Ning, H. Zhang, H. Yang, S.J. Chang, A generalized Faustmann model with multiple carbon pools, Forest Policy Econ. 169 (2024) 103363.

[44] Z. Zhang, M. Lin, D. Li, R. Wu, R. Lin, C. Yang, An AUV-Enabled Dockable Platform for Long-Term Dynamic and Static Monitoring of Marine Pastures, IEEE J. Ocean. Eng. (2024).

[45] H. Lv, J. Zeng, Z. Zhu, S. Dong, W. Li, Study o… distribution and structural performance of heptagonal six-fiv… t alte… cable dome with inner hole, in: Structures, Vol. 65, Elsevi… 24, p. 1067…

[46] G. Tian, Y. Sun, S. Liu, C. Li, S. Chen, … X. Zhang, Y. L… Li, H. Shang, Therapeutic effects of wenxin keli in cardio… r diseases: a… perimental and mechanism overview, Front. Pha… ol. 9 (2… 005.

[47] B.-P. Zhu, Z. Tang, L.-H. Zhao… Wang, C.-Z… Yin… Yu, W.-F. Tang, R. Xiong, J. Shi, Synthesis o… [Ti2] O4 by spark … tering, Mater. Lett. 61 (2007) 578–581.

[48] X. Sui, H. Gao, Y. Sun, Q… G. … frared super-resolution imaging method based on retina mi… moti… d Phys. T… l. 60 (2013) 340–345.

[49] Z.L. Zhao LiChun… Liu Ying… Wang Z… Tang Nong, L.J. Leng Jing, Z. B. Zheng Bing… Liu YingYing… Li… Liquid Chromatography/Mass Spectrome… s and hepatopr… effect of steamed Platycodi Radix on acute al… -ind… iver injury, (20…).

[50] X. Sui, Q. Chen, G. … uniformity correction of infrared images based on infrared radiation and … g time of thermal imager, Optik 124 (2013) …

[51] … Yu, Z. Ning, W.-Y. Chang, J. Chang, H. Yang, Optimal harvest decisions for the management of carbon sequestration forests under price uncertainty and risk preferences, … Policy Econ. 151 (2023) 102957.

[52] … X. Sui, Q. Che… Gu, X. Shen, Infrared super-resolution imaging based on …mpressed se… g, Infrared Phys. Technol. 63 (2014) 119–124.

[53] … …g, C… Z. Wu, W. Dong, Y. Wang, A. Mian, Hyperrectangle Embedding fo… 3D Scene Graph Prediction From RGB Sequences, IEEE Trans. Pattern Anal. Mach. Intell. (2025).

[54] … Chen, W. Zhou, H. Kuang, Z. Chen, J. Yang, Z. Chen, F. Chen, Dynamic model …ibration of rack vehicle on curve line, Veh. Syst. Dyn. (2025) 1–19.

[55] H.H. Zhang, J.B. Chao, Y.W. Wang, Y. Liu, H.M. Yao, Z.P. Zhao, K. Niu, 5G base station antenna array with heatsink radome, IEEE Trans. Antennas Propag. 72 (2024) 2270–2278.

[56] B. Zhu, N.Y. Chan, J. Dai, K.K. Shung, S. Takeuchi, Q. Zhou, New fabrication of high-frequency (100-MHz) ultrasound PZT film kerfless linear array [Correspondence], IEEE Trans. Ultrason. Ferroelectr. Freq. Control 60 (2013) 854–857.

[57] J.-Y. Yang, Y.-T. Zhong, W.-N. Hao, X.-X. Liu, Q. Shen, Y.-F. Li, S. Ren, Z. Wang, W. Li, L.-C. Zhao, Retracted Article: The PI3K/Akt and NF-κB signaling pathways are involved in the protective effects of Lithocarpus polystachyus (sweet tea) on APAP-induced oxidative stress injury in mice, RSC Adv. 10 (2020) 18044–18053.

[58] F.J. Huang, Z.Z. Wu, From "unusual sequences" to human diseases, in: Medicine Sciences and Bioengineering: Proceedings of the 2014 International Conference on Medicine Sciences and Bioengineering (ICMSB2014), Kunming, Yunnan, China, August 16-17, 2014, CRC Press, 2015, p. 443.

[59] G. Tian, S. Zhao, X. Wang, X. Tang, Y. Wang, Q. Deng, X. Wang, S. Li, Y. Tang, C. Liu, Leg-bathing in decoction in treating rheumatism, experimented on rats and demonstrated by visualization of knee joint synovium, Acta Medica Mediterranea 32 (2016) 1873–1879.

[60] G. Tian, C. Wu, J. Li, B. Liang, F. Zhang, X. Fan, Z. Li, Y. Wang, Z. Li, D. Liu, Network pharmacology based investigation into the effect and mechanism of Modified Sijunzi Decoction against the subtypes of chronic atrophic gastritis, Pharmacol. Res. 144 (2019) 158–166.

[61] M. Ortúzar, M. Esterhuizen, D.R. Olicón-Hernández, J. González-López, E. Aranda, Pharmaceutical pollution in aquatic environments: a concise review of environmental impacts and bioremediation systems, Front. Microbiol. 13 (2022) 869332.

[62] A. Kulkarni, S.E. Miller, Chapter 2 – Analysis of pharmaceuticals in the environment, in: M. Török (Ed.), Contemporary Chemical Approaches for Green and Sustainable Drugs, Elsevier, 2022, pp. 27–45.

[63] C. Meiqun, Z. Gui, S. Kehuan, W. Zhengzhi, Application of bioinformatics technique to screen of differentially expressed proteins in serums of breast cancer patients with white and yellow coating, in: 2011 IEEE International Symposium on IT in Medicine and Education, Vol. 1, IEEE, 2011, pp. 216–220.

[64] J. Sheng, H. Yang, Linking water markets with payments for watershed services: the eastern route of China's South-North Water Transfer Project, Agric. Water Manage. 295 (2024) 108733.

[65] J. Sheng, H. Yang, Collaborative models and uncertain water quality in payments for watershed services: China's Jiuzhou River eco-compensation, Ecosyst. Serv. 70 (2024) 101671.

[66] X. Sui, Q. Chen, G. Gu, N. Liu, Multi-sampling and filtering technology of IRFPA, Optik 122 (2011) 1037–1041.

RETRACTED

[67] B. Bai, J. Chen, B. Zhang, Flowing-water remediation simulation experiments of lead-contaminated soil using UCB technology, Int. J. Phytorem. 27 (2025) 761–770.

[68] G. Tian, C. Zhao, X. Zhang, W. Mu, Y. Jiang, X. Wei, M. Zhao, Z. Shi, Y. Jin, J. Si, Evidence-based traditional Chinese medicine research: two decades of development, its impact, and breakthrough, J. Evidence-Based Med. 14 (2021) 65–74.

[69] Z.-Q. Yang, K.-P. Hou, Y. Cheng, B. Yang, Study of column-hemispherical penetration grouting mechanism based on power-law fluid, Chin. J. Rock Mech. Eng. 33 (2014) 3840–3846.

[70] J. Zhai, W. Mu, J. Si, Y. Li, C. Zhao, H. Shang, H. Li, G. Tian, Acupuncture for constipation in patients with stroke: protocol of a systematic review and *meta*-analysis, BMJ Open 8 (2018) e020400.

[71] Z. Yang, J. Xiong, X. Zhao, X. Meng, S. Wang, R. Li, Y. Wang, M. Chen, N. He, Y. Yang, Column-hemispherical penetration grouting mechanism for Newtonian fluid considering the tortuosity of porous media, Processes 11 (2023) 1737.

[72] Z. Yang, L. Wei, Y. Liu, N. He, J. Zhang, H. Xu, Discussion on the relationship between debris flow provenance particle characteristics, gully slope, and debris flow types along the karakoram highway, Sustainability 15 (2023) 5998.

[73] X. Li, Q. Dai, Z. Shi, H. Chen, Y. Hu, X. Wang, X. Zhang, G. Tian, Clinical efficacy and safety of electroacupuncture in migraine treatment: a systematic review and network *meta*-analysis, Am. J. Chin. Med. 47 (2019) 1755–1780.

[74] X.-Y. Li, S. Lin, Y. Lin, L.-Q. Huang, X.-M. Xie, G.-H. Tian, Medication rules of traditional Chinese medicine compounds for pain, Zhongguo Zhong Yao Za Zhi (2023) 3386–3393.

[75] S. Xiu-Bao, C. Qian, G. Guo-Hua, L. Ning, Research on the response model of microbolometer, Chin. Phys. B 19 (2010) 108702.

[76] Z.Q. Yang, Y.Y. Zhu, D.S. Zou, L.P. Liao, Activity degree evaluation of glacial debris flow along international Karakorum Highway (KKH) based on fuzzy theory, Adv. Mat. Res. 261 (2011) 1167–1171.

[77] J. Gao, Friction coefficient estimation of the clutch in automatic transmission based on improved persistent excitation condition, in: 2017 Chinese Automation Congress (CAC), IEEE, 2017, p. 5647.

[78] Z.Q. Yang, Y.Y. Zhu, D.S. Zou, Formation conditions and risk evaluation of glacial debris flow disasters along International Karakorum Highway (KKH), Italian J. Eng. Geol. Environ. (2011) 1031–1037.

[79] Y. Zhiquan, N. Xiangdong, H. Kepeng, L. Wei, G. Yanhui, Relationships between water-cement ratio and rheological characteristics of power-law cement grouts, Electron. J. Geotech. Eng. 20 (2015) 5825–5831.

[80] T.B. Henry, J.W. Kwon, K.L. Armbrust, M.C. Black, Acute and chronic toxicity of five selective serotonin reuptake inhibitors in Ceriodaphnia dubia, Environ. Toxicol. Chem. 23 (2004) 2229–2233.

[81] P.H. Roberts, K.V. Thomas, The occurrence of selected pharmaceuticals in wastewater effluent and surface waters of the lower Tyne catchment, Sci. Total Environ. 356 (2006) 143–153.

[82] Y. Xiao, Y. Yang, D. Ye, J. Zhang, Quantitative precision se… tempora… transformation based pose control for spacecraft proximi… …IEEE Trans. Aerosp. Electron. Syst. (2024).

[83] Z. Ni, J. Ma, Y. Liu, B. Li, A. Nazarov, H. Li, Z. Yuan, …g, X. Wa… …umerical Analysis of Ultrasonic Spot Welding of Cu/Cu Joints, …er. … (2025) 1–12.

[84] Z.Z. Wu, J.G. Wang, X.L. Zhang, M.Q. Cao, …liminary stu… …aliva proteomics of different furs in digestive … diseases, in: 2… …EE International Symposium on IT in Me… …ducation, Vol. …E, 2009, pp. 56–61.

[85] D. Ye, B. Wang, L. Wu, E.A. Del…o-Chanona, Z. … …O-SRPP: A decentralized pivoting path planning metho… self-reconfigurab… …llites, IEEE Trans. Ind. Electron. 71 (2024) 14318… …7.

[86] Y. Xiao, Y. Yang, D. Ye, S…g-transf…tion based attitude tracking control for rigid spacecraft with …bed ti… …d prescribed bound, IEEE Trans. Aerosp. Electron. Syst. (2024).

[87] C. Yang, Y. Chen, W. Sun, …, M. Diao, …n, Extreme soil salinity reduces N and P met… …nd rela… …crobi… …work complexity and community immigrati… …e, … …n. Res. … …120361.

[88] B. Bai, …ang, Y. …Y. Zong, A …odynamic multi-field model for unsa… …d sulfate… …ne soils consi…ring crystallization process, Comput. Geote… …l (2…

[89] J. Du, H. …ang, W. Bao, …Li, Solidification microstructure reconstruction and its effe… …hase transformation, grain boundary transformation mechanism, a… …hanical properties of TC4 alloy welded joint, Metall. Mater. Trans. A 55 (202… …3–1206.

[90] W. Bao, H. Liu, F. W…ng, J. Du, Y. Wang, H. Li, X. Ye, Keyhole critical failure criteria and variation rule under different thicknesses and multiple materials in K-TIG welding, J. Manuf. Process. 126 (2024) 48–59.

[91] D. Zheng, X. Cao, Provably efficient service function chain embedding and protection in edge networks, IEEE/ACM Trans. Networking (2024).

[92] B. Zhu, W. Wei, Y. Li, X. Yang, Q. Zhou, K.K. Shung, KNN-based single crystal high frequency transducer for intravascular photoacoustic imaging, in: 2017 IEEE International Ultrasonics Symposium (IUS), IEEE, 2017, pp. 1–4.

[93] B. Bai, H. Wu, Q. Nie, J. Liu, X. Jia, Granular Thermodynamic Migration Model Suitable for High-Alkalinity Red Mud Filtrates and Test Verification, Int. J. Numer. Anal. Meth. Geomech. (2025).

[94] B. Zhu, Z. Zhang, T. Ma, X. Yang, Y. Li, K.K. Shung, Q. Zhou, (100)-Textured KNN-based thick film with enhanced piezoelectric property for intravascular ultrasound imaging, Appl. Phys. Lett. 106 (2015).

[95] B. Zhu, J. Xu, Y. Li, T. Wang, K. Xiong, C. Lee, X. Yang, M. Shiiba, S. Takeuchi, Q. Zhou, Micro-particle manipulation by single beam acoustic tweezers based on hydrothermal PZT thick film, AIP Adv. 6 (2016).

[96] L. Zhao, P. Yu, T. Yang, G. Zhou, N. Tang, Inhibitory Effect of Semen Litchi Drug Serum on the Proliferation of Human Hepatoma HepG2 Cells and Expression of VEGF and MMP-9, J. Coll. Physicians Surg. Pak. 29 (2019) 532–536.

[97] G. Tian, K. Qian, X. Li, M. Sun, H. Jiang, W. Qiu, X. Xie, Z. Zhao, L. Huang, S. Luo, Can a holistic view facilitate the development of intelligent traditional Chinese medicine? A survey, IEEE Trans. Comput. Soc. Syst. 10 (2023) 700–713.

[98] J. Huang, Z. Yan, P. Qiu, Y. Mo, Q. Cao, Q. Li, L. Huo, L. Zhao, A new coumarin-acridone compound as a fluorescence probe for Fe3+ and its application in living cells and zebrafish, Molecules 26 (2021) 2115.

[99] Z. Yang, X. Zhao, M. Chen, J. Zhang, Y. Yang, W. Chen, X. Bai, M. Wang, Q. Wu, Characteristics, dynamic analyses and hazard assessment of debris flows in Niumiangou Valley of Wenchuan County, Appl. Sci. 13 (2023) 1161.

[100] Y. Wu, Y. Lin, M. Zhang, K. He, G. Tian, Causal association between circulating inflammatory markers and sciatica developmen… …ian randomization study, Front. Neurol. 15 (2024) 1380719.

[101] Y.-Y. Lin, H. Xia, G.-Y. Xia, X.-Y. Li, C. Zha… …T. Zhang, S… …Z.-H. Tian, Effect of Siegesbeckiae Herba in treating chro… …n, Zhongguo Z… …yao za zhi= Zhongguo Zhongyao Zazhi= China Journa… …inese Materia… …dica 45 (2020) 1851–1858.

[102] J. Yang, Analysis of clinical m… …tion charact… …of … …y turmeric oil injection based on real wor… …a, Chin… …N Drugs … …547–552.

[103] T. Yue, W. Zhang, H. Pei… …anzeng… …, J. Yang, …ao, Z. Zhang, S. Xiong, X. Yang, Monascus pigm… …otect… …ne marrow-derived stem cells for heart failure treatment, B… …st. M… …2024) 270…

[104] Y. Hou, K. Tang… …ang, D. … …Zhang… …tative mating on blood type: Evidence from… …million Chines… …na… …Proc. Natl. Acad. Sci. 119 (2022) e2209643…

[105] L. Zhao… …ong… …ng, H. Gao, N… …ng, Effect of Astragalus polysaccharide on the expression of… …and EGFR in mice with Lewis transplantable lung cancer, J. Coll. Physici… …rg. Pak. 29 (2019) 392–394.

[106] …Q. Zhang, J. Lia… …i, X. Tan, N. Tang, Astrocyte elevated gene-1 …duces autophagy in diabe… …cardiomyopathy through upregulation of KLF4, J. Cell. Biochem. 120 (2019) 9709–9715.

[1… …Z.-Z. Wu, J.-G… …ng, X.-L. Zhang, Diagnostic model of saliva protein finger print …nalysis of pat… …with gastric cancer, World J. Gastroenterol.: WJG 15 (2009) …5.

[108] …X. Li… …uang, X. Xie, T. Luo, G. Tian, Acupuncture combined with Ch… …medicine for discogenic low back pain: protocol for a multicentre, randomised controlled trial, BMJ Open 14 (2024) e088898.

…Li, S. Lin, Y. Lin, Y. Su, C. Wang, L. Huang, J. Zhao, G. Tian, The analgesic …anism of Xi Shao Formula research on neuropathic pain based on metabolomics, Journal of Traditional Chinese Medical Sciences 10 (2023) 448–460.

[110] Q.-Q. Dai, X.-Y. Li, G.-Y. Xia, H. Xia, X.-H. Wei, H.-C. Shang, S. Lin, G.-H. Tian, Research progress on animal models of chronic pain and its application in traditional Chinese medicine research, Zhongguo Zhong yao za zhi= Zhongguo Zhongyao Zazhi= China J. Chin. Materia Medica 45 (2020) 5866–5876.

[111] Y. Huang, T. Hong, X. Chang, T. Ma, Travel and Regional Development: A Quantitative Analysis of China, J. Reg. Sci. (2025).

[112] Z. Yang, D. Zhang, C. Li, Z. Zhang, Y. Zhu, Y. Yang, N. He, X. Bai, W. Xi, D. He, Column penetration and diffusion mechanism of Bingham fluid considering displacement effect, Appl. Sci. 12 (2022) 5362.

[113] Z. Yang, X. Fan, Y. Yang, K. Hou, J. Du, X. Chen, Y. Mi, C. Jiang, J. Zhang, Y. Guo, Deformation patterns and failure mechanism of high and steep stratified rock slopes with upper steep and lower gentle style induced by step-by-step excavations, Environ. Earth Sci. 81 (2022) 229.

[114] Z. Yang, Y. Liu, M. Chen, X. Wang, C. Ye, X. Li, W. Chen, Y. Yang, B. Wang, C. Li, Influence of coupling effects between gravel soil porosity and cement grout weight on diffusion laws and morphologies of penetration grouting, Appl. Sci. 12 (2022) 7601.

[115] Z. Yang, Y. Ding, Y. Yang, Y. Zhu, J. Zhang, Y. Guo, X. Chen, Hydration-time-dependent rheological behaviors of Newtonian cement grouts with different water cement ratios, Trans. Chin. Soc. Agric. Eng 36 (2020) 161–167.

[116] B. Bai, H. Wu, R. Zhou, N. Wu, B. Zhang, A granular thermodynamic framework-based coupled multiphase-substance flow model considering temperature driving effect, J. Rock Mech. Geotech. Eng. (2024).

[117] J. Gao, T. Shen, Online calibration of spark advance for combustion phase control of gasoline SI engines, in: 2016 Seventh International Conference on Intelligent Control and Information Processing (ICICIP), IEEE, 2016, pp. 185–190.

[118] G. Huang, J. Liang, X. Chen, J. Lin, J. Wei, D. Huang, Y. Zhou, Z. Sun, L. Zhao, Isolation and Identification of Chemical Constituents from Zhideke Granules by Ultra-Performance Liquid Chromatography Coupled with Mass Spectrometry, J. Anal. Methods Chem. 2020 (2020) 8889607.

[119] W. Jiang, Q. Ma, English teaching system design based on ID3 algorithm and node optimization, PeerJ Comput. Sci. 9 (2023) e1486.

[120] J. Gao, T. Shen, Cylinder pressure sensor-based real-time combustion phase control approach for SI engines, IEEJ Trans. Electr. Electron. Eng. 12 (2017) 244–250.

[121] J. Gao, Z. Liu, Integrated powertrain control of gearshift for automatic transmission, in: 2010 First International Conference on Pervasive Computing, Signal Processing and Applications, IEEE, 2010, pp. 239–242.

[122] B. Bai, F. Bai, Q. Nie, X. Jia, A high-strength red mud–fly ash geopolymer and the implications of curing temperature, Powder Technol. 416 (2023) 118242.

RETRACTED

[123] T. Zhang, S. Xu, W. Zhang, New approach to feedback stabilization of linear discrete time-varying stochastic systems, IEEE Trans. Autom. Control (2024).
[124] G.-H. Tian, S.-S. Tao, M.-T. Chen, Y.-S. Li, Y.-P. Li, H.-C. Shang, X.-Y. Tang, J.-X. Chen, H.-B. Tang, Electroacupuncture treatment alleviates central poststroke pain by inhibiting brain neuronal apoptosis and aberrant astrocyte activation, Neural Plast. 2016 (2016) 1437148.
[125] L.-C. Zhao, W. Li, Y.-F. He, X.-H. Li, Z. Wang, W.-C. Liu, Y.-N. Zheng, J. Liang, Fast and Sensitive LC-DAD-ESI/MS Method for Analysis of Saikosaponins c, a, and d from the Roots of Bupleurum Falcatum (Sandaochaihu), Molecules 16 (2011) 1533–1543.
[126] J. Gao, Z. Liu, Adaptive estimator for the friction characteristics of the clutch in automatic transmission, in: Proceedings of the 29th Chinese Control Conference, IEEE, 2010, pp. 6004–6009.
[127] C. Meiqun, G. Zifan, S. Kehuan, W. Zhengzhi, Application of iTRAQ quantitative proteomics in identification of serum biomarkers in breast cancer, in: 2011 4th International Conference on Biomedical Engineering and Informatics (BMEI), IEEE, 2011, pp. 1658–1663.
[128] J. Gao, Z. Liu, Static output feedback control of the linear system with parameter uncertainties, in: 2010 8th World Congress on Intelligent Control and Automation, IEEE, 2010, pp. 503–508.
[129] J. Gao, Y. Wu, T. Shen, Experimental comparisons of hypothesis test and moving average based combustion phase controllers, ISA Trans. 65 (2016) 504–515.
[130] L.-C. Zhao, J. Liang, W. Li, K.-M. Cheng, X. Xia, X. Deng, G.-L. Yang, The use of response surface methodology to optimize the ultrasound-assisted extraction of five anthraquinones from Rheum palmatum L, Molecules 16 (2011) 5928–5937.
[131] W. Zhengzhi, W. Jiguo, Z. Xiaoli, C. Meiqun, W. Anmin, Preliminary studies on diagnostic cast of peptic ulcer based on saliva proteome and bioinformatics, in: 2011 4th International Conference on Biomedical Engineering and Informatics (BMEI), IEEE, 2011, pp. 1597–1600.
[132] X. Sui, Q. Chen, G. Gu, Micro-scanning system using flat optics for resolution improvement of infrared images, Optik 124 (2013) 2292–2297.
[133] Z. Wu, M. Li, Y. Li, Y. Zhang, X. Jia, M. Chen, Quantitative study of tiantai I on superoxidative dismutase and lipofuscin in relevant cerebral areas of spontaneous Alzheimer disease in mice, Chinese J. Tissue Eng. Res. (2005) 178–181.
[134] Z.Z. Wu, Y.H. Li, M. Li, J.G. Wang, M. Yang, M.Y. Chen, Effect of Natural Brain-Kenetine on variance of Gene expression profiles in MSC and hippocampus of AD rats analyzed by Gene chips and bioinformatics techniques, in: 2008 IEEE International Symposium on IT in Medicine and Education, IEEE, 2008, pp. 730–737.
[135] Z. Wu, M. Li, Y. Zhang, M. Chen, Study on relationship between the thickness of tongue fur and the expressions of apoptosis-related genes of the tongue epi[illegible] cells in patients with diseases of the digestive system, J. Traditional Chin. [illegible] Chung i tsa Chih Ying wen pan 27 (2007) 148–152.
[136] J. Zima, I. Švancara, J. Barek, K. Vytřas, Recent advances in electroanalysis [illegible] organic compounds at carbon paste electrodes, Crit. Rev. Anal. Chem. 39 (200[illegible] 204–227.
[137] W. Shang, C. Wang, Z. Wu, Study of Hollow Nanoneedle[illegible] Performance by Nanorobotic Manipulation System, in: [illegible] IEEE 1[illegible] International Conference on Nano/Micro Engineered [illegible] Molecul[illegible]stems (NEMS), IEEE, 2019, pp. 366–369.
[138] I. Švancara, K. Vytřas, J. Barek, J. Zima, Carb[illegible]ste e[illegible]n modern electroanalysis, Crit. Rev. Anal. Chem. 31 ([illegible] 311–345.
[139] S. Yamada, H. Sato, Some physical prope[illegible] glassy carbon, [illegible]193 (1962) 261–262.
[140] G. Jenkins, K. Kawamura, Structure [illegible]lassy [illegible] Nature 231 (1[illegible]71) 175–176.
[141] H. Zittel, F. Miller, A Glassy-C[illegible]n Electrode for V[illegible]metry, Anal. Chem. 37 (1965) 200–203.
[142] M. Sheraz, X.-F. Sun, A[illegible]qui, Y. [illegible], S. Hu, R. Sun, Cellulose-Based Electrochemical Sens[illegible]nsors 2[illegible]25) 645.
[143] A.R. Harris, Current pe[illegible]he safe el[illegible]trical stimulation of peripheral nerves with platinum elec[illegible]bioelectro[illegible]d. 3 (2020) 37–49.
[144] M. Zamani, C[illegible]nerich, [illegible]urst, P[illegible]t advances in gold electrode fabricatio[illegible]ow[illegible]urce set[illegible]ng, Lab Chip 23 (2023) 1410–1419.
[145] M. Sop[illegible]ous, J.K[illegible]kinson, A [illegible] of screen-printed silver/silver chloride (Ag/[illegible] referenc[illegible]ctrodes potentially suitable for environmental potent[illegible]ic s[illegible]tors, A 267 (2017) 106–120.
[146] J. Gao, Y[illegible]hen, On-[illegible]statistical combustion phase optimization and control of S[illegible]ine engines, Appl. Therm. Eng. 112 (2017) 1396–1407.
[147] Z. Wu, M. Li, [illegible]Y. Zhang, Influence of Tiantai No. 1 Recipe on learning and memory function[illegible]ntaneous Alzheimer disease models, Chin. J. Tissue Eng. Res. (2005) 180–18[illegible].
[148] Z. Wu, M. Li, A.C. Huang, X. Jia, Y. Li, M. Chen, Effects of serum containing natural cerebrolysin on glucose-regulated protein 78 and CCAAT enhancer-binding protein homologous protein expression in neuronal PC12 cells following tunicamycin-induced endoplasmic reticulum stress, Neural Regen. Res. 4 (2009) 92–97.
[149] Z. Wu, C. Meiqun, J. Wang, X. Zhang, A. Wu, Study on saliva proteome and bioinformatics in patients with chronic gastritis, in: 2011 IEEE International Symposium on IT in Medicine and Education, Vol. 1, IEEE, 2011, pp. 221–225.
[150] Z.Z. Wu, J.G. Wang, X.L. Zhang, M.Q. Cao, Preliminary study of salivary proteome on the diagnosis for discrimination of gastric cancer and chronic gastritis, in: 2009 IEEE International Symposium on IT in Medicine and Education, Vol. 1, IEEE, 2009, pp. 70–75.
[151] R. Su, H. Tang, F. Xi, Sensitive electrochemical detection of p-nitrophenol by pre-activated glassy carbon electrode integrated with silica nanochannel array film, Front. Chem. 10 (2022) 954748.
[152] J. Lenik, J. Nieszporek, Construction of a glassy carbon ibuprofen electrode modified with multi-walled carbon nanotubes and cyclodextrins, Sens. Actuators B 255 (2018) 2282–2289.
[153] Z. Wu, Y. Li, A. Huang, M. Li, M. Yang, M. Chen, Natural cerebrolysin induces neuronal differentiation in bone marrow mesenchymal stem cells, Neural Regen. Res. 4 (2009) 178–185.
[154] Z.Z. Wu, Y.H. Li, A.C. Huang, M. Li, X.L. Zhang, J.G. Wang, M. Yang, M.Y. Chen, Endoplasmic reticulum stress induced by tunicamycin and antagonistic effect of Tiantai No. 1 (天泰 1 号) on mesenchymal stem cells, Chinese journal of integrative medicine 16 (2010) 41–49.
[155] Z.-Z. Wu, Effects of Tiantai No. 1 on relative neuropeptides of spontaneous aged dementia mice, Chin. J. Neurosci. 20 (2004) 167–170.
[156] Z. Wu, M. Li, Y. Li, X. Jia, Y. Zhang, Effects of Tiantai I on the activity of central cholinergic system in mice with spontaneous A[illegible]ease, Chin. J. Tissue Eng. Res. (2006) 163–165.
[157] W. ZZ, A.C. Huang, J. de Vellis, L. YH, Ef[illegible]f Tiantai N[illegible]a beta-amyloid-induced neurotoxicity and NF-kappa B [illegible]MP responsive[illegible]ent-binding protein, Chinese Journal of Integrative Me[illegible]14 (4) (2008[illegible]6–292.
[158] D. Xiang, D. He, H. Sun, P. Gao, [illegible]ng, J. [illegible]ICMPE-N[illegible]n unsupervised network for underwater image[illegible]ation with h[illegible]aram[illegible]estimation based on homology constraint, Op[illegible]er Technol. 186 ([illegible]316.
[159] M. Feng, C. Yan, Z. Wu, [illegible]ng, Y. W[illegible]A. Mian, [illegible]y-Enhanced 3D Scene Graph Reasoning from R[illegible]Sequ[illegible], IEEE Trans. Circuits Syst. Video Technol. (2025).
[160] M. Li, L.-J. Zhan[illegible]H. Tian, [illegible]Hao, Ne[illegible]ating repeated chest CT in COVID-19 pn[illegible]nia, J. Formos[illegible]A[illegible]119 (2020) 993.
[161] Y. Yu, F. Z[illegible]n, H. Fujita, J. [illegible]ng, E. Chen, CrowdFPN: crowd counting[illegible]scal[illegible]nced and locat[illegible]aware feature pyramid network, Appl. Intell. 55 ([illegible]2025) 1–[illegible].
[162] Y. Guan, Z. Cui, W. Zh[illegible]construction in off-axis digital holography based on [illegible]ustering and the[illegible]nal Fourier transform, Opt. Laser Technol. 186 ([illegible]2025) 112622.
[163] M.-X. Zhang, J.-F. Tang, D.-T. Cai, S. Li, K. Li, S.-Q. Chi, G.-Q. Cao, X. Zhang, Y. Zhou, Z.-B[illegible] Technical performance, surgical workload and patient [illegible]utcomes of r[illegible]c and laparoscopic surgery for pediatric choledochal cyst: a [illegible]lticenter re[illegible]pective cohort and propensity score-matched study, [illegible]bili[illegible]rgery and Nutrition (2025).
[164] M[illegible]Zhang, S. Chi, X. Chang, J. Zeng, H. Bian, G. Cao, S. Li, Y. Zhou, L. Rong, Robotic-assisted proctosigmoidectomy versus laparoscopic-assisted [illegible]ave pull-through for hirschsprung disease: Medium-term outcomes from a [illegible]pective multicenter study, Ann. Surg. 281 (2025) 689–697.
[165] N. Singh, U. Sanyal, J.L. Fulton, O.Y. Gutiérrez, J.A. Lercher, C.T. Campbell, Quantifying adsorption of organic molecules on platinum in aqueous phase by hydrogen site blocking and in situ X-ray absorption spectroscopy, ACS Catal. 9 (2019) 6869–6881.
[166] X. Chen, X. Cheng, J.J. Gooding, Detection of trace nitroaromatic isomers using indium tin oxide electrodes modified using β-cyclodextrin and silver nanoparticles, Anal. Chem. 84 (2012) 8557–8563.
[167] M. Pimpilova, A brief review on methods and materials for electrode modification: electroanalytical applications towards biologically relevant compounds, Discover Electrochemistry 1 (2024) 12.
[168] F. Qindong, Y. Zhang, G. Wei, X. Huang, Coupling coordination analysis of urban social vulnerability and human activity intensity, Environ. Res. Commun. 7 (2025) 035009.
[169] F. Qindong, L. Qiaosu, Y. Xiaoyu, Spatiotemporal assessment of recreation ecosystem service flow from green spaces in Zhengzhou's main urban area, Humanities and Social Sciences Communications 12 (2025) 1–14.
[170] H. Yang, X. Zhang, A rethinking of the production approach in IPCC: its objectiveness in China, Sustainability 8 (2016) 216.
[171] G. Yang, L. Wang, J. Jia, D. Zhou, D. Li, Chemically modified glassy carbon electrode for electrochemical sensing paracetamol in acidic solution, J. Solid State Electrochem. 16 (2012) 2967–2977.
[172] H. Yang, T. Yuan, X. Zhang, S. Li, A decade trend of total factor productivity of key state-owned forestry enterprises in China, Forests 7 (2016) 97.
[173] H. Yang, Y. Nie, C. Ji, Study on China's timber resource shortage and import structure: natural forest protection program outlook, 1998 to 2008, Forest Products Journal 60 (2010) 408–414.
[174] H. Yang, X. Li, Potential variation in opportunity cost estimates for REDD+ and its causes, Forest Policy Econ. 95 (2018) 138–146.
[175] L.-N. Ding, Y.-H. Hu, T. Li, M. Li, Y.-T. Li, Y.-Z. Wu, J. Cao, X.-L. Tan, A GDSL motif-containing lipase modulates Sclerotinia sclerotiorum resistance in Brassica napus, Plant Physiol. 196 (2024) 2973–2988.
[176] Y. Hongqiang, C. Ji, Y. Nie, H. Yinxing, China's wood furniture manufacturing industry: industrial cluster and export competitiveness, For. Prod. J. 62 (2012) 214–221.
[177] J. Xu, L. Fan, C. Chen, G. Lu, B. Li, T. Tu, Study on fuel injection stability improvement in marine low-speed dual-fuel engines, Appl. Therm. Eng. 253 (2024) 123729.
[178] A. Geng, J. Chen, H. Yang, Assessing the greenhouse gas mitigation potential of harvested wood products substitution in China, Environ. Sci. Tech. 53 (2019) 1732–1740.

RETRACTED

[179] D. Zhang, B. Li, Y. Wei, H. Zhang, G. Lu, L. Fan, J. Xu, Investigation of injection and flow characteristics in an electronic injector featuring a novel control valve, Energ. Conver. Manage. 327 (2025) 119609.
[180] X. Li, X. Zhang, H. Yang, Estimating the opportunity costs of avoiding oil palm-based deforestation in Indonesia: Implications for REDD+, Chin. J. Population, Resour. Environ. 18 (2020) 9–15.
[181] A. Geng, H. Zhang, H. Yang, Greenhouse gas reduction and cost efficiency of using wood flooring as an alternative to ceramic tile: A case study in China, J. Clean. Prod. 166 (2017) 438–448.
[182] J. Sheng, R. Ding, H. Yang, Corporate green innovation in an aging population: Evidence from Chinese listed companies, Technol. Forecast. Soc. Chang. 202 (2024) 123307.
[183] Y. Nie, C. Ji, H. Yang, The forest ecological footprint distribution of Chinese log imports, Forest Policy Econ. 12 (2010) 231–235.
[184] S. Malinowski, C. Wardak, J. Jaroszyńska-Wolińska, P.A.F. Herbert, R. Panek, Cold Plasma as an Innovative Construction Method of Voltammetric Biosensor Based on Laccase, in, Sensors (2018).
[185] H. Liao, H. Ma, H. Meng, N. Kang, L. Wang, Ropinirole suppresses LPS-induced periodontal inflammation by inhibiting the NAT10 in an ac4C-dependent manner, BMC Oral Health 24 (2024) 510.
[186] C. Wang, L. Yang, M. Hu, Y. Wang, Z. Zhao, On-demand airport slot management: tree-structured capacity profile and coadapted fire-break setting and slot allocation, Transportmetrica A: Transport Sci. (2024) 1–35.
[187] X. Qin, W. Yang, Z. Zhang, V.W. Wangari, Simulation and design of T-shaped barrier tops including periodic split ring resonator arrays for increased noise reduction, Appl. Acoust. 236 (2025) 110751.
[188] L. Feng, Q. Xue, F. Liu, Q. Cao, J. Feng, L. Yang, F. Zhang, Voltammetric determination of ofloxacin by using a laser-modified carbon glassy electrode, Microchim. Acta 187 (2020) 86.
[189] F. Qian, R. Jia, M. Cheng, A. Chaudhary, S. Melhi, S.D. Mekkey, N. Zhu, C. Wang, F. Razak, X. Xu, An overview of polylactic acid (PLA) nanocomposites for sensors, Adv. Compos. Hybrid Mater. 7 (2024) 75.
[190] T. Peng, Z. Ning, H. Yang, Embodied CO2 in China's trade of harvested wood products based on an MRIO model, Ecol. Ind. 137 (2022) 108742.
[191] X. Li, Z. Ning, H. Yang, A review of the relationship between China's key forestry ecology projects and carbon market under carbon neutrality, Trees, Forests and People 9 (2022) 100311.
[192] B. Zhu, D. Wu, Q. Zhou, J. Shi, K. Shung, Lead zirconate titanate thick film with enhanced electrical properties for high frequency transducer applications, Appl. Phys. Lett. 93 (2008).
[193] Z. Su, B. Liu, Y. Zhang, J. Song, B. Qian, W. Liu, S. Sun, J. Qiu, Y. Dai, Pr… processing of Nb-Si alloy via water-jet guided laser: Realization of inhibite… oxidation and small-taper, Opt. Laser Technol. 187 (2025) 112853.
[194] S. Zhang, Y. Yang, H. Yang, A meshless symplectic algorithm for nonlinear w… equation using highly accurate RBFs quasi-interpolation, Appl. Math Comput. 3… (2017) 110–120.
[195] Q. Long, H. Yan, H. Wu, S. Qiu, X. Zhou, T. Qiu, Influence … leaching agent anions on the leaching of aluminium impurities … c-type … earth ores: A DFT simulation combined with experimental … cation, … Purif. Technol. 354 (2025) 128768.
[196] B. Zhu, Q. Zhou, J. Shi, K. Shung, S. Irisawa, … arated hydrothermal lead zirconate titanate thick … or high fr… transducer applications, Appl. Phys. Lett. 94 (2009).
[197] N. An, T. Chen, J. Zhang, G. Wang, M. … g, Rational El… emical Design of Cuprous Oxide Hierarchic… croar… ures and Their Derivatives for SERS Sensing Applications, … all Methods 8 … 2300910.
[198] H. Yang, X. Zhang, Y. Hong, … ication, product… d carbon stock of harvested wood products i… a, BioResources 9 (20… 311–4322.
[199] Z. Ni, J. Ma, A. Nazarov, … an, X. W… S. Ao, H. Ji, P. Guo, J. Qin, Improving the weldability and m… ical pr… y of ultrasonic spot welding of Cu sheets through a surface gradi… J. Mater. … s. Technol. 36 (2025) 2652–2668.
[200] B. Zhu, Y. Zh… , J. O… X. Y… Li, W. Wei, New potassium sodium ni… sin… rystal w… -independent high-performance for photoa… c angio… hy of athe… otic lesion, Sci. Rep. 6 (2016) 39679.
[201] J. Li… n, Simul… ous determina… on of dopamine and serotonin on gold nanoc… ove… de composite modified glassy carbon electrode… tuators B … (2007) 486–493.
[202] J. Wei, J.-B… Q. Cao, Y.-W. Zhu, Y. Wang, G.-P. Hang, Enhanced sensing of ascorbic acid, … ine and serotonin at solid carbon paste electrode with a nonionic polyme… Talanta 83 (2010) 190–196.
[203] M. Choudhary, S. Si… al, D. Nandi, K. Mallick, Single step synthesis of gold–amino acid composite, with the evidence of the catalytic hydrogen atom transfer (HAT) reaction, for the electrochemical recognition of Serotonin, Physica E 77 (2016) 72–80.
[204] C. Xue, X. Wang, W. Zhu, Q. Han, C. Zhu, J. Hong, X. Zhou, H. Jiang, Electrochemical serotonin sensing interface based on double-layered membrane of reduced graphene oxide/polyaniline nanocomposites and molecularly imprinted polymers embedded with gold nanoparticles, Sens. Actuators B 196 (2014) 57–63.
[205] N.K. Sadanandhan, M. Cheriyathuchenaaramvalli, S.J. Devaki, A. R. Ravindranatha Menon, PEDOT-reduced graphene oxide-silver hybrid nanocomposite modified transducer for the detection of serotonin, J. Electroanal. Chem. 794 (2017) 244–253.
[206] M. Mazloum-Ardakani, A. Khoshroo, Electrocatalytic properties of functionalized carbon nanotubes with titanium dioxide and benzofuran derivative/ionic liquid for simultaneous determination of isoproterenol and serotonin, Electrochim. Acta 130 (2014) 634–641.
[207] G. Ran, C. Chen, C. Gu, Serotonin sensor based on a glassy carbon electrode modified with multiwalled carbon nanotubes, chitosan and poly (p-aminobenzenesulfonate), Microchim. Acta 182 (2015) 1323–1328.
[208] B. Dinesh, V. Veeramani, S.-M. Chen, R. Saraswathi, In situ electrochemical synthesis of reduced graphene oxide-cobalt oxide nanocomposite modified electrode for selective sensing of depression biomarker in the presence of ascorbic acid and dopamine, J. Electroanal. Chem. 786 (2017) 169–176.
[209] W. Zhang, N. Xu, N. Zhao, A.A. Al-Barakati, Adaptive neural finite-time self-triggered control for nonstrict-feedback nonlinear systems with sensor faults, Rob. Intell. Autom. (2025).
[210] M. Pimpilova, A brief review on methods and materials for electrode modification: electroanalytical applications towards biologically relevant compounds, Discover, Electrochemistry 1 (2024) 1–20.
[211] D. Cai, Z. Yang, R. Tong, H. Huang, C. Zhang, Y. Xia, Binder-Free MOF-Based and MOF-Derived Nanoarrays for Flexible Electroch… gy Storage: Progress and Perspectives, Small 20 (2024) 2305778.
[212] J. Zhao, N. Xu, B. Niu, X. Zhao, A. Alorfi, … namic event… red optimal control for stochastic interconnected n… r systems with … hed disturbances via adaptive dynamic progra… J. Franklin … 362 (2025) 107360.
[213] Z. Gu, S. Yan, C.K. Ahn, D. Yu, … Xie, Event-t… d di… ve tracking control of networked control… ms with distribut… ication delay, IEEE Syst. J. 16 (2021) 3320–…
[214] F. Wei, N. Xu, X. Zhao, … Al… kati, Dynamic memory event-triggered adaptive neural pr… bed… ite conse… control for high-order MASs with privacy pr… tion, Co… Nonlin… Numer. Simul. 145 (2025) 108693.
[215] M. Liu, N. … ng, G. Zong, X… Li, Hierarchical non-singular termina… ng … ontrol for cons… ned under-actuated nonlinear systems against s… sor fault… linear Dyn. (2025) 1–17.
[216] B.-R. Adhikari, M. Gov… A. Chen, Carbon Nanomaterials Based … emical Sensors/… ors for the Sensitive Detection of Pharmaceutical … Biological Compounds, … nsors (2015) 22490–22508.
[217] … E. Elanthamilan, S.-F. Wang, Integration of NiCoFe layered double hydroxide@ graphitic carb… itride composite: An electrochemical sensing platform for … cetaminophe… g in water and urine samples, J. Environ. Chem. Eng. 13 …025) 116928.
[218] … han… nar, R.A. Khan, K. Ahmad, H. Kim, Fabrication of Sulfur-Doped R… phene Oxide Modified Glassy Carbon Electrode (S@rGO/GCE) Based Acetaminophen Sensor, Inorganics (2022).
… Anbumannan, K. Suresh, Sensitive and selective electrochemical detection of … minophen based on GO-MnO2-modified electrode, Carbon Lett. 35 (2025) 243–255.
[220] C.R. Vanoni, R.B. Goularte, A.R. Silva Lima, N.B. Guedes, M.R. Scheide, G. C. Bazzo, R.L.T. Parreira, G.F. Caramori, G.R. Nagurniak, M.E. Rosso Dotto, H. K. Stulzer, C.L. Jost, An alumina-modified glassy carbon electrode: a robust platform for accurate nimodipine detection in pharmaceutical applications, Anal. Methods 17 (2025) 2203–2213.
[221] R. Liu, B. Wu, R. Liu, X. Zhang, Y. Sun, J. Ma, Development of Ag/ZIF-67/g-C3N4/GCE electrochemical sensor to detect chloramphenicol with the assistance of machine learning, Microchem. J. 212 (2025) 113510.
[222] Ç.A. Özata, N. Erk, W. Bouali, A.A. Genc, H.E. Hassan Ahmed, A.A. Matin, M. Soylak, Electrochemical detection of riociguat using COF-PTA@CuCoFe-LDH modified glassy carbon electrode: A novel approach for pharmaceutical and biological analysis, Microchem. J. 211 (2025) 113158.
[223] L. Qian, A.R. Thiruppathi, R. Elmahdy, J. van der Zalm, A. Chen, Graphene-Oxide-Based Electrochemical Sensors for the Sensitive Detection of Pharmaceutical Drug Naproxen, Sensors (2020).
[224] M. Ahmadi-Kashani, H. Dehghani, A new multifunctional electrocatalyst based on PbS nanostructures decorated with graphene/polyaniline-modified glassy carbon electrode for selective detection of non-steroidal anti-inflammatory drug naproxen, Microchem. J. 200 (2024) 110320.
[225] S. Ansari, M.S. Ansari, H. Devnani, S.P. Satsangee, R. Jain, CeO2/g-C3N4 nanocomposite: A perspective for electrochemical sensing of anti-depressant drug, Sens. Actuators B 273 (2018) 1226–1236.
[226] Y. Wang, S. Li, Y. Gao, B. Du, S. Vafaei, M. Li, H. Wu, X. Tong, Y. Chen, Synthesis of poly (L-cysteine)/g-C3N4 modified glassy carbon electrodes for electrochemical detection of methotrexate as a medicine for treatment of breast cancer in pharmaceutical fluid samples, Chemosphere 331 (2023) 138769.
[227] G. Venkata Prasad, V. Vinothkumar, S. Joo Jang, D. Eun Oh, T. Hyun Kim, Multi-walled carbon nanotube/graphene oxide/poly(threonine) composite electrode for boosting electrochemical detection of paracetamol in biological samples, Microchem. J. 184 (2023) 108205.
[228] G. Muungani, W.E. van Zyl, A CaCuSi 4O 10/GCE electrochemical sensor for detection of norfloxacin in pharmaceutical formulations, RSC Adv. 13 (2023) 12799–12808.
[229] I. Rus, A. Pusta, M. Tertiș, C. Barbălată, I. Tomuță, R. Săndulescu, C. Cristea, Gemcitabine Direct Electrochemical Detection from Pharmaceutical Formulations Using a Boron-Doped Diamond Electrode, Pharmaceuticals (2021).
[230] A.R.A. Gowda, M.B. Kavyashree, M.A. Sangamesha, M.B. Shivaswamy, S. Jeevitha, H.P. Shivaraju, H.S.N. Prasad, The rGO@S-CeMgZn nanocomposite: new insights into the selective and simultaneous electrochemical detection of

RETRACTED

ciprofloxacin and acetaminophen in biological fluids, New J. Chem. 49 (2025) 7342–7357.
[231] A. Yari, Development of a porous polycitric acid-branch coated graphene oxide nanocomposite infused with Fe3O4 nanoparticles for advanced electrochemical sensing of piroxicam in authentic and pharmaceutical samples, Microchem. J. 214 (2025) 114078.
[232] F. Garkani Nejad, H. Beitollahi, I. Sheikhshoaie, A UiO-66-NH2 MOF/PAMAM Dendrimer Nanocomposite for Electrochemical Detection of Tramadol in the Presence of Acetaminophen in Pharmaceutical Formulations, Biosensors (2023).

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